Contamination Bias-Free Estimation and Inference via Debiased Machine Learning

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Abstract

Regression studies with several treatment arms—or with a continuous dosage split into bins—often report coefficients that are contaminated: each estimate mixes the arm's own causal effect with spill-overs from the other arms. We provide a single, closed-form influence function that removes this bias for both discrete and continuous treatments. Inserted into cross-fitted double machine learning estimators, the score delivers $\sqrt{n}$–consistent and semiparametrically efficient inference under high-dimensional or non-linear first-stage models. A data-driven trimming rule handles weak overlap. Under independent clusters, we provide a cross-fit-by-cluster scheme and a cluster-summed variance estimator that deliver valid cluster-robust inference. The new estimator nests the three existing fixes of Goldsmith-Pinkham *et al.* (2024) and supports a growing number of arms. For any fixed arm we obtain root-n consistent and efficient inference even when the total number of arms $K = K_n$ grows with n; for joint Gaussian inference across *all* K_n coordinates we prove a many–arm CLT when $K_n = o(n^{1/3})$ under uniform fourth-moment bounds. Monte-Carlo experiments show up to 50% reductions in RMSE when outcomes are non-linear or overlap is weak. An open-source `Python` package, `dmlContam`, implements the full workflow.

JEL Codes: C14, C21, C55

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1 Introduction

Causal studies routinely evaluate many mutually-exclusive policy arms (e.g. school-reopening tiers, tax brackets) or *use* continuous scores that act as dosages. In a recent paper, (Goldsmith-Pinkham *et al.*, 2024) show that regressing outcomes on a saturated set of arm dummies or on dosage bins yields coefficients contaminated by cross-arm effects unless the regression function is perfectly linear. While their three formal fixes (EW, CW, IA) mitigate the problem, they retain the assumptions of linear conditional means and strong overlap.

The contamination issue sits at the junction of several literatures. Early work on multi-valued treatments treated each arm separately (Imbens, 2000; Imai and van Dyk, 2004), or relied on multinomial propensity scores (Li *et al.*, 2018; Leite *et al.*, 2019; Li and Li, 2019; Collier *et al.*, 2021). Continuous-dose analogues go back to (Hirano and Imbens, 2004; Imai and Ratkovic, 2014) and, more recently, the semiparametric and nonparametric influence-function approach of Kennedy (2022, 2024). Orthogonal scores coupled with double/debiased machine learning (DML) have proved powerful for binary treatments and high-dimensional covariates (Chernozhukov *et al.*, 2018; Athey and Wager, 2018; Farrell *et al.*, 2021; Kennedy, 2024), but no unified orthogonal score existed for multi-arm or continuous designs, nor for settings with weak overlap. The latter has generated its own line of research on optimal trimming and overlap weights (Crump *et al.*, 2009; Lopez and Gutman, 2017; Kallus and Oprescu, 2023).

This paper provides a general solution. We provide efficient scores for contamination-free estimation, unify discrete and continuous designs, and embed them in a cross-fitted DML framework that algebraically eliminates contamination for *both* discrete treatments $D \in \{0, \ldots, K\}$ and continuous doses $D \in \mathbb{R}$. Plugging that score into the cross-fitted DML framework of Chernozhukov *et al.* (2018) yields root-n, semiparametrically efficient estimators even when the nuisance functions $m(D, W) = E[Y \mid D, W]$ and $p_k(W)$ are high-dimensional and highly non-linear. Weak overlap is handled by a data-driven trimming rule that minimizes an estimated mean squared error, adapting the ideas of Crump *et al.* (2009) and Kallus and Oprescu (2023).

Main contributions are as follows. A *single* orthogonal score that *unifies and strictly extends* the three GPHK estimators (IA, EW, CW) and nests the classical AIPW score when $K = 1$. First contamination-free framework for *continuous* treatments and a *many-arm* extension. For any fixed arm k (not depending on n), the trimmed cross-fitted DML estimator admits standard root-n inference without restricting how the total number of arms $K = K_n$ may grow (beyond uniform moment/overlap conditions). For *joint* inference on $(\theta_1, \ldots, \theta_{K_n})$, we prove a many-arm CLT when $K_n = o(n^{1/3})$ under uniform fourth-moment bounds on the scores (Corollary 6.2). We also provide a cross-fit-by-cluster scheme with a cluster-summed variance estimator for valid cluster-robust inference (Theorem 6.3).

Beyond the strands already mentioned, our results connect to (i) the semiparametric efficiency literature (Newey, 1990; Chamberlain, 1992; Hahn, 1998; Graham, 2011) for multi-valued treatments, (ii) recent advances in automatic orthogonalisation for continuous doses (Kennedy, 2024; Lewis and Syrgkanis, 2021), and (iii) high-dimensional central-limit theorems (Chernozhukov *et al.*, 2017) that justify inference when K diverges. We build on these but provide the first contamination-robust, root-n-regular estimator that applies across discrete and continuous designs.

Section 2 introduces setup and assumptions. Section 3 derives the EIFs. The estimator and trimming rule appear in Section 4; asymptotic theory in Section 6. Section 7 designs simulations and reports the evidence from Monte Carlo simulations. Section **??** revisits a GPHK application and adds a continuous-dose example. Section 8 concludes. Technical proofs are in Appendix A.

2 Setup and Assumptions

The goal of this section is two-fold. First, we spell out the stochastic environment—what variables are observed and what causal quantities are of interest. Second, we collect the high-level conditions under which the subsequent orthogonal-score construction and asymptotic theory will operate. Throughout we let $Z_i = (Y_i, D_i, W_i)$ denote the observable outcome (Y), treatment (D), and covariates (W) for unit $i = 1, \ldots, n$, and assume the draws are i.i.d. for notational simplicity. Two treatment regimes are covered.

We begin with the canonical setting of $K+1$ mutually exclusive arms. Formally, $D_i \in \{0, 1, \ldots, K\}$, where $k = 0$ is the control arm, and we write $D_{ik} = 1\{D_i = k\}$ for the associated dummies. The *propensity score* for arm k is

$$p_k(W) \;=\; P(D_i = k \mid W_i), \qquad k = 0, \ldots, K,$$

so $p_k(W) \in (0,1)$ for each k under our overlap conditions below.

Throughout the paper we rely on a standard selection on observables condition.

Assumption 2.1 (Unconfoundedness—Discrete)**.** *The vector of potential outcomes* $\{Y_i(0), Y_i(1), \ldots, Y_i(K)$ *is independent of the realised treatment* D_i *conditional on covariates* W_i*:*

$$\{Y_i(0), Y_i(1), \ldots, Y_i(K)\} \;\perp\!\!\!\perp\; D_i \,\big|\, W_i.$$

Interpretation. After conditioning on W_i—which may be high-dimensional—assignment to each arm is "as good as random," ruling out unobserved confounders. This is the discrete analogue of the continuous unconfoundedness assumption used below.

Our primary target is the arm-specific average treatment effect (ATE) relative to control,

$$\theta_k \;=\; E\big[\,Y_i(k) - Y_i(0)\big], \qquad k = 1, \ldots, K. \tag{1}$$

Point identification of θ_k follows directly from Assumption 2.1 and the law of iterated expectations.

Many empirical studies record a dosage or index rather than a finite set of arms—think of tax rates, pollution levels, or test scores. We therefore let D_i take values in a compact interval $\mathcal{D} \subset \mathbb{R}$ and denote by $f(d \mid W)$ the conditional density of the dose. The conditional mean function is $m(d, W) = E[Y \mid D = d, W]$.

Assumption 2.2 (Unconfoundedness—Continuous)**.** *For every $d \in \mathcal{D}$ the potential outcome $Y_i(d)$ is independent of the realised dose D_i conditional on covariates W_i:*

$$Y_i(d) \;\perp\!\!\!\perp\; D_i \,\big|\, W_i.$$

In the continuous setting we focus on the average *partial* effect (APE),

$$\theta \;=\; E\big[\partial_d m(D_i, W_i)\big],$$

i.e. the expected marginal effect of a small dose change evaluated at the observed D_i.

Assumption 2.3 (Smoothness)**.** *The regression function $m(d, W)$ and the log-density $\log f(d \mid W)$ are twice continuously differentiable in d and lie in a Hölder class of order $\alpha > 2$ with respect to W.*

These regularity conditions guarantee that the efficient influence function we derive in Section 3.1 exists and, importantly, that local-polynomial estimators can attain the $n^{-1/4}$ convergence rate that orthogonal scores require.

Estimation becomes unstable when $p_k(W)$ or $f(D \mid W)$ approach zero. Rather than assume a uniform lower bound, we allow the support violations to vanish slowly with the sample size and control them via trimming.

Assumption 2.4 (Weak overlap (both designs))**.** *Let $\tau_n \downarrow 0$ be a deterministic trimming threshold. There exists $\delta > 0$ such that*

$$P\big(\min_k p_k(W) < \tau_n\big) = O(\tau_n^{\delta}) \quad \textit{(discrete)} \quad \textit{or} \quad P\big(f(D \mid W) < \tau_n\big) = O(\tau_n^{\delta}) \quad \textit{(continuous).}$$

Remark 2.1 (Trimming indicators and shorthand)**.** *We use the following trimming indicators throughout:*

$$T_i^{disc}(\tau) = \mathbf{1}\left\{\min_{\ell \le K} p_\ell(W_i) \ge \tau\right\}, \qquad T_i^{cont}(\tau) = \mathbf{1}\{f(D_i \mid W_i) \ge \tau\}. \tag{2}$$

When the design is clear from context we write $T_i(\tau)$ for either (2)*.*

The tails of the propensity or density are allowed to get thinner as the sample grows, but not so fast that we lose $\sqrt{n}$ information after trimming the worst-overlap observations. The adaptive rule of Section 6.2 chooses τ_n on a data-dependent grid.

For the accuracy of machine-learning first stages, all subsequent estimators rely on non-parametric fits $\hat{m}, \hat{p}_k, \hat{f}$. The next assumption imposes the minimal rate conditions under which Neyman orthogonality can partial out first-stage error.

Assumption 2.5 (Anti-concentration near the trimming threshold)**.** *There exist $c > 0$ and $\kappa \in (0, 2]$ such that, uniformly over $\tau \in \mathcal{G}_n$ and all $h > 0$ small enough,*

$$\Pr\Big(|\min_{\ell \le K} p_\ell(W) - \tau| \le h\Big) \;\le\; c\, h^\kappa \quad \textit{(discrete)}, \qquad \Pr\Big(|f(D \mid W) - \tau| \le h\Big) \;\le\; c\, h^\kappa \quad \textit{(continuous)}.$$

That is, the distribution of the trimming margin has a bounded local density uniformly over the grid.

Remark 2.2. *Assumption 2.5 is implied, for example, if the distribution of $\min_{\ell \le K} p_\ell(W)$ (respectively $f(D \mid W)$) admits a bounded Hölder-continuous density in a neighborhood of each $\tau \in \mathcal{G}_n$. It is used only to control indicator disagreements $\Pr(\hat{T}_i(\tau) \ne T_i(\tau))$ when we combine hard trimming with estimated propensities/densities.*

Assumption 2.6 (First-stage ML rates)**.** *On the* trimmed *support,*

$$\|\hat{m} - m\|_{2,T} = o_p(n^{-1/4}), \qquad \|\hat{p}_k - p_k\|_{2,T} = o_p(n^{-1/4}),$$

and, for continuous doses, $\|\partial_d \hat{m} - \partial_d m\|_{2,T}$, $\|\hat{f} - f\|_{2,T} = o_p(n^{-1/4})$.

Honest causal forests, boosted trees, deep nets, and local-polynomial estimators all satisfy these rates under either sparsity or smoothness—see Appendix A.12 for citations.

We finally record two technical conditions that are invisible in classical low-dimensional proofs but become essential once we allow (i) clusters of heterogeneous size, and (ii) a growing number of arms K_n.

Remark 2.3 (Trimmed L_2 norm)**.** *For any scalar function $g(Z)$ and threshold τ, we define*

$$\|g\|_{2,T(\tau)}^2 := \frac{E[g(Z)^2\,\mathbf{1}\{T(\tau)=1\}]}{P(T(\tau)=1)}.$$

All rate statements in Assumption 2.6 are with respect to $\|\cdot\|_{2,T(\tau)}$.

Assumption 2.7 (Uniform moments and overlap margin as K grows)**.** *Let $\varphi_k^\star$ denote the efficient score in (4). There exists a trimming lower bound $\underline{\tau}_n \in (0,1)$ (the minimum of the grid in Section 6.2) such that for all large n:*

(i) Uniform overlap tail: $P(\min_\ell p_\ell(W) < t) \le C\,t^\delta$ *for all $t \in (0,1)$ and some $\delta > 0$;*

(ii) Uniform outcome fourth moments: $\sup_{\ell \le K_n} E\big[\,|Y - m_\ell(W)|^4\,\big] < \infty$*;*

(iii) Joint control of fourth moments and K_n:

$$\sup_{k \le K_n} E\Big[\varphi_k^\star(Z)^4\,\mathbf{1}\{\min_\ell p_\ell(W) \ge \underline{\tau}_n\}\Big] \le C_4\,\underline{\tau}_n^{-q}$$

for some finite C_4 and $q \in [2,4]$; and the growth/trimming coupling

$$\frac{K_n^3\,\underline{\tau}_n^{-q}}{n} \;\longrightarrow\; 0.$$

The last display ensures the Lyapunov ratio in the many-arm CLT vanishes even if the fourth moments inflate as $\underline{\tau}_n \downarrow 0$. When a non-vanishing grid is used (fixed $\underline{\tau} > 0$), it holds automatically.

Remark 2.4 (How to pick the grid in practice)**.** *If you wish to let the trimming grid vanish with n while K_n grows, set its minimum according to $\underline{\tau}_n \asymp (K_n^3/n)^{1/q}$ for the q implied by your learners (e.g. $q = 2$ under standard tail behavior). This meets Assumption 2.7 and keeps the many-arm CLT valid.*

Remark 2.5 (Fixed-arm vs. growing-K_n scope)**.** *Unless explicitly stated otherwise (see Corollary 6.2), all asymptotic statements in the paper are* fixed-arm*: the index k is held fixed as $n \to \infty$. No restriction on how the total number of arms $K = K_n$ may grow is needed for these fixed-arm results beyond the uniform moment condition $\sup_{k \le K_n} E[\varphi_k^\star(Z)^4] < \infty$. The growth condition $K_n = o(n^{1/3})$ is used only for the* joint *many-arm CLT.*

Under Assumption 2.3 a trimmed L_2 rate of $o_p(n^{-1/4})$ is attainable for local-polynomial density and regression estimators whenever the treatment dimension $d \le 4$ Cattaneo et al., 2024.

Remark 2.6. *Detailed moment and cross-fit regularity proofs are collected in Appendix A.6.*

3 Orthogonal Influence Functions

Contamination arises because a saturated regression forces the coefficients on different treatment arms (or bins) to *compete for the same residual variation.* The cure is an *orthogonal* influence function that: (i) identifies the causal parameter in the usual semiparametric sense, (ii) removes every linear dependence on the nuisance functions $\eta(\cdot)$, and therefore (iii) remains first-order valid even when η is estimated by a flexible learner. The two influence functions below meet all three goals and reduce to classical AIPW scores when $K = 1$.

The following influence functions both identify the parameters and algebraically remove the contamination bias of Goldsmith-Pinkham *et al.* (2024).

3.1 Continuous Treatment

Let $f(d \mid W)$ be the conditional density of the dose and write $s(d, W) = \partial_d \log f(d \mid W)$ for its score. Conditional outcome regressions are denoted $m(d, W) = E[Y \mid D = d, W]$.

The efficient influence function (EIF) is

$$\varphi^{(c)}(Z; \theta, \eta) \;=\; s(D, W)\{Y - m(D, W)\} + \partial_d m(D, W) - \theta, \qquad \eta = (m, f). \tag{3}$$

The first term multiplies the outcome residual by the density score— the analogue of a "clever covariate" familiar from binary AIPW estimators—while the $\partial_d m$ term shifts the moment so that its expectation is exactly the average partial effect θ. Any first-order error in $\hat{m}$ or $\hat{f}$ cancels by construction, which is the essence of Neyman orthogonality proven below.

Lemma 3.1 (Orthogonality—Continuous)**.** *Under Assumptions 2.2 and 2.3, and the boundary condition in Assumption 3.1 below, the influence function* $\varphi^{(c)}$ (i) *uniquely identifies* θ *and* (ii) *is Neyman-orthogonal with respect to the nuisance bundle* $\eta = (m, f)$*.*

Proof. The proof proceeds in two parts. First, we show that the expectation of the score is zero at the true parameter values, which establishes identification of $\theta = E[\partial_d m(D, W)]$. Second, we compute the pathwise derivative of the moment condition with respect to perturbations in the nuisance functions m and f and show it is zero. This Neyman orthogonality property is key to the $\sqrt{n}$-consistency of the DML estimator, as it renders the estimator insensitive to first-order errors in the estimation of the nuisance functions. The full, detailed proof is provided in Appendix A.2. □

Assumption 3.1 (Boundary control)**.** *Either (a) D is monotonically transformed to full real-line support so that $h_m(d, W) f(d \mid W) \to 0$ at $\pm\infty$; or (b) boundary-corrected local* *polynomial estimators are used for m and $\log f$ so that $h_m(d, W)f(d \mid W) \to 0$ at the finite endpoints of $\mathcal{D}$.*

Remark 3.1 (Learners and rates). *Under Assumption 3.1 and the smoothness conditions used for density/local polynomial estimators, the trimmed L_2 rates required for our orthogonal score hold in low dimensions (e.g., $d \le 4$), and the integration-by-parts step is valid. State explicitly whether $\partial_d m$ is produced by the same local polynomial fit as m (preferred) or by a separate fit.*

Remark 3.2 (Vector-valued doses). *Assumption 3.1 is imposed* coordinate-wise *when $D \in \mathbb{R}^d$, i.e. the boundary product $h_m(\boldsymbol{d}, W)f(\boldsymbol{d} \mid W)$ must vanish at every face of the rectangular support $\mathcal{D} \subset \mathbb{R}^d$. All proofs that rely on integration by parts therefore extend verbatim to $d > 1$.*

Remark 3.3 (Rates under boundary-robust estimation). *Under Assumption 2.3 with $\alpha > 2$ and treatment dimension $d \le 4$, boundary-corrected local polynomials deliver trimmed L_2 rates $o_p(n^{-1/4})$ for $(m, \partial_d m)$ and $(f, \partial_d \log f)$ on any grid value $\tau > 0$, which suffices for Neyman-orthogonal DML. See Cattaneo et al. (2024) and references therein.*

Remark 3.4 (Square-Integrability of the Score $s(D, W)$). *The validity of the asymptotic theory requires the influence function to have a finite second moment, which in turn depends on the score function $s(D, W) = \partial_d \log f(D|W)$ being square-integrable. Assumption 2.3 states that $\log f(d|W)$ is twice continuously differentiable, which implies that both $s(d, W)$ and its derivative $\partial_d s(d, W)$ are continuous functions. On the trimmed support, where $f(D|W) \ge \tau_n > 0$, the denominator of $s(D, W) = (\partial_d f)/f$ is bounded away from zero. As continuous functions on a compact domain are bounded, both $\partial_d f(d, W)$ and $f(d|W)$ (on the trimmed set) are bounded. Therefore, $s(D, W)$ is bounded on the trimmed support. A bounded random variable on a probability space necessarily has finite moments of all orders, ensuring that $E[s(D, W)^2 \cdot \mathbf{1}\{f(D|W) \ge \tau_n\}] < \infty$.*

Plugging $\varphi^{(c)}(Z_i; \hat{\theta}, \hat{\eta})$ into a cross-fitted DML estimator therefore yields $\sqrt{n}$-consistent and semiparametrically efficient inference for continuous treatments—even when m and f are fitted by, say, boosted trees or deep nets.

3.2 Discrete Multi-Arm Treatment

Set-up. For mutually exclusive arms $D \in \{0, 1, \dots, K\}$ let $m_\ell(W) = E[Y \mid D = \ell, W]$ and $p_\ell(W) = P(D = \ell \mid W)$. Contamination in a plain dummy regression stems from the fact that the dummies sum to one; the EIF below corrects this algebraically by subtracting the control term arm-by-arm.

The efficient, contamination-free influence function for arm k is

$$\begin{aligned}\varphi_k^\star(Z;\theta_k,\eta) = &\big[m_k(W) - m_0(W)\big] \\ &+ \frac{D_{ik}}{p_k(W)}\big\{Y - m_k(W)\big\} - \frac{D_{i0}}{p_0(W)}\big\{Y - m_0(W)\big\} - \theta_k,\end{aligned} \tag{4}$$

with nuisance bundle $\eta = (m_0, \ldots, m_K, p_0, \ldots, p_K)$.
The structure mirrors an AIPW score for a *binary* treatment, except that the outcome-regression and IPW corrections are stacked in a "treatment-minus-control" fashion.

Lemma 3.2 (Identification & Neyman orthogonality)**.** *Let $\tau_k(W) = E[Y_i(k) - Y_i(0) \mid W]$. Then $E[\varphi_k^\star] = E[\tau_k(W)] - \theta_k$; hence $E[\varphi_k^\star] = 0$ iff $\theta_k = E[Y_i(k) - Y_i(0)]$. Moreover, $\partial_r E[\varphi_k^\star(Z;\theta_k,\eta + rh)]_{r=0} = 0$ for every mean-zero perturbation $h = (h_m, h_p)$, and $\varphi_k^\star$ attains the semiparametric efficiency bound.*

Because $\varphi_k^\star$ is also **doubly robust**, an empiricist only needs one of the two first-stage models—propensities or outcome regressions—to be well estimated. This is crucial when flexible learners are applied in high-dimensional W.

Proof. The proof establishes two key properties. First, identification, which shows that the score has a unique root at the true ATE θ_k. Second, Neyman orthogonality, which demonstrates that the moment condition is locally insensitive to errors in the nuisance functions η. This latter property is essential for the validity of the DML estimator. The logic relies on the law of iterated expectations and the specific structure of the score, which balances regression-adjustment and inverse-propensity weighting terms. A full, step-by-step derivation is provided in Appendix A.4. □

Remark 3.5 (Double robustness)**.** *The moment condition in Lemma 3.2 continues to hold if either all outcome regressions (m_ℓ) or all propensity scores (p_ℓ) are consistently estimated, as shown in Appendix A.4.*

3.3 Root-n CLT for the Orthogonal Score under Deterministic Trimming

The next result shows that a *fixed* threshold τ_n already delivers $\sqrt{n}$ inference when weak overlap is modestly severe.

Theorem 3.1 (Asymptotic normality under deterministic trimming)**.** *Let $\hat\theta_k$ be the DML estimator for the arm-k ATE from Algorithm 1, and let a similar definition apply to the continuous-case APE $\hat\theta$. Under Assumptions 2.1–2.6 and the following conditions :*

1. *The nuisance estimators converge at the required rate:* $\|\hat{m}_\ell - m_\ell\|_{2,T} = o_p(n^{-1/4})$ *and* $\|\hat{p}_\ell - p_\ell\|_{2,T} = o_p(n^{-1/4})$ *(and similarly for the continuous case).*

2. *The trimming sequence satisfies* $n\tau_n^{2\delta} \to 0$ *(with* δ *from Assumption 2.4).*

3. *The conditional average treatment effects,* $\tau_k(W)$*, are bounded.*

4. *The variance of the influence function,* $V_k = E[\varphi_k^\star(Z)^2]$*, is finite and positive.*

Then the DML estimator is consistent, asymptotically normal, and semiparametrically efficient:

$$\sqrt{n}(\hat{\theta}_k - \theta_k) \xrightarrow{d} N(0, V_k).$$

Furthermore, the centered plug-in variance estimator

$$\hat{V}_k \;=\; \Bigl(\sum_i T_i\Bigr)^{-2} \sum_{i=1}^{n} T_i \left\{\hat{\varphi}_{ik} - \bar{\varphi}_{T,k}\right\}^2, \qquad \bar{\varphi}_{T,k} \;=\; \Bigl(\sum_i T_i\Bigr)^{-1} \sum_{i=1}^{n} T_i\, \hat{\varphi}_{ik},$$

is consistent for $V_k = E[\varphi_k^\star(Z)^2]$*.*

Proof sketch. Decompose $\sqrt{n}(\hat{\theta}_k - \theta_k)$ into the trimmed sample average of the influence function, a cross-fit remainder, and a bias term. Neyman orthogonality makes the remainder $o_p(1)$ once the nuisance estimates obey the $n^{-1/4}$ rate, while the bias is $O\bigl(\sqrt{n}\,\tau_n^{\delta}\bigr) = o(1)$ because $n\tau_n^{2\delta} \to 0$. A Lyapunov CLT applied to the leading term then yields the stated limit. Full details are in Appendix A.5. Since $\tau_n \downarrow 0$ and $n\tau_n^{2\delta} \to 0$, we have $\bar{\varphi}_{T,k} = \bigl(\sum_i T_i\bigr)^{-1} \sum_i T_i\, \hat{\varphi}_{ik} = o_p(1)$ by the linearization in (14) and Lemma A.3. Hence the centered and uncentered trimmed second moments differ by $o_p(1)$, so either form consistently estimates V_k under deterministic vanishing trimming. □

4 Estimator & Implementation

Remark 4.1. *Uniform moment and cross-fit regularity proofs for Algorithm 1 are given in Appendix A.6.*

Remark 4.2. *Clustered sampling is handled formally in Section 6.4. All inference statements below admit a cluster-robust version under the assumptions recorded there.*

When the nuisance fits are restricted as in § 5, Algorithm 1 reproduces IA, EW, or CW exactly. If observations are grouped in G clusters $\{\mathcal{C}_g\}_{g=1}^{G}$ with arbitrary within-cluster

Algorithm 1 Cross-Fitted DML with Trimming

Split *clusters* into J folds $\{\mathcal{I}_j\}_{j=1}^{J}$ and form the corresponding hold-out observation sets $\bigcup_{g\in\mathcal{I}_j}\mathcal{C}_g$.
for $j=1$ **to** J **do**
    Estimate nuisances on the complement $\bigcup_{m\neq j}\mathcal{I}_m$: $\hat{m}^{(-j)}$, $\hat{p}^{(-j)}$ *(plus* $\partial_d\hat{m}^{(-j)}, \hat{f}^{(-j)}$ *if* D *is continuous).*
    for $i \in \bigcup_{g\in\mathcal{I}_j}\mathcal{C}_g$ **do**
        Trimming indicator $T_i(\hat{\tau}) = \mathbf{1}\{\min_k \hat{p}_k^{(-j)}(W_i) \geq \hat{\tau}\}$ *or* $T_i(\hat{\tau}) = \mathbf{1}\{\hat{f}^{(-j)}(D_i \mid W_i) \geq \hat{\tau}\}$.
        Compute raw moment (no $-\theta$):

$$\hat{\psi}_i = \begin{cases} \hat{s}(D_i,W_i)\{Y_i - \hat{m}(D_i,W_i)\} + \partial_d\hat{m}(D_i,W_i), & \text{((3) without } -\theta) \\ \big[\hat{m}_k(W_i) - \hat{m}_0(W_i)\big] + \dfrac{D_{ik}}{\hat{p}_k(W_i)}\{Y_i - \hat{m}_k(W_i)\} - \dfrac{D_{i0}}{\hat{p}_0(W_i)}\{Y_i - \hat{m}_0(W_i)\}, & \text{((4) without } -\theta_k) \end{cases}$$

where $\hat{s}(d,w) = \partial_d \log \hat{f}(d \mid w)$.
        Store trimmed raw moment $\tilde{\psi}_i = T_i(\hat{\tau})\,\hat{\psi}_i$.
    end for
end for
Point estimate: $\hat{\theta} = \Big(\sum_i T_i(\hat{\tau})\Big)^{-1}\sum_i \tilde{\psi}_i$.
Empirical influence values: $\hat{\varphi}_i = \hat{\psi}_i - \hat{\theta}$, $\tilde{\varphi}_i = T_i(\hat{\tau})\,\hat{\varphi}_i$.
Variance: $\hat{V} = \Big(\sum_i T_i(\hat{\tau})\Big)^{-2}\sum_i T_i(\hat{\tau})\,\big\{\hat{\varphi}_i - \bar{\varphi}_T\big\}^2$, where $\bar{\varphi}_T = \Big(\sum_i T_i(\hat{\tau})\Big)^{-1}\sum_i T_i(\hat{\tau})\hat{\varphi}_i$.

dependence and independence across clusters, *form folds by clusters*: $\{\mathcal{I}_j\}_{j=1}^{J}$ is a partition of the cluster set $\{1,\ldots,G\}$ and the estimation/validation uses $\bigcup_{g\in\mathcal{I}_j}\mathcal{C}_g$ as the jth hold-out block. All nuisance fits and trimming thresholds for units in $\bigcup_{g\in\mathcal{I}_j}\mathcal{C}_g$ are trained on $\bigcup_{m\neq j}\bigcup_{g\in\mathcal{I}_m}\mathcal{C}_g$.

5 Mappings to ATE(IA), CW, and EW

This section makes the algebra explicit: Interacted–ATE (IA) and Common–Weight (CW) are plug-in instances of our unified orthogonal score, while EW coincides with the pairwise AIPW score on the binary subsample $\{D \in \{0,k\}\}$.

5.1 Interacted–ATE (IA): explicit mapping in sample

Let the saturated linear model be

$$Y_i = \alpha_0 + \beta_0^\top W_i + \sum_{\ell=1}^{K} D_{i\ell}\,(\alpha_\ell + \beta_\ell^\top W_i) + \varepsilon_i, \qquad E[\varepsilon_i \mid D_i, W_i] = 0,$$

and let $\hat m_\ell(W) = \hat\alpha_\ell + \hat\beta_\ell^\top W$ be the OLS fits on the training folds (cross-fitted). Define the arm-specific OLS residuals $r_{i\ell} = Y_i - \hat m_\ell(W_i)$ and note the normal equations

$$\sum_{i\in\mathcal{I}_j} \mathbf{1}\{D_i = \ell\}\, r_{i\ell} = 0, \qquad \sum_{i\in\mathcal{I}_j} \mathbf{1}\{D_i = \ell\}\, r_{i\ell} W_i = 0, \quad (\ell = 0,\ldots,K). \tag{5}$$

Our efficient score for arm k (cf. (4)) is

$$\varphi_k^\star(Z_i;\theta_k,\eta) = \big[m_k(W_i) - m_0(W_i)\big] + \frac{D_{ik}}{p_k(W_i)}\{Y_i - m_k(W_i)\} - \frac{D_{i0}}{p_0(W_i)}\{Y_i - m_0(W_i)\} - \theta_k.$$

Evaluate it with nuisances $(\hat m, \hat p)$ and drop the $-\theta_k$ term to form the raw moment

$$\hat\psi_{ik} = \big[\hat m_k(W_i) - \hat m_0(W_i)\big] + \frac{D_{ik}}{\hat p_k(W_i)}\, r_{ik} - \frac{D_{i0}}{\hat p_0(W_i)}\, r_{i0}.$$

Add and subtract $D_{ik}\, r_{ik}$ and $D_{i0}\, r_{i0}$ to re-express the IPW residual terms:

$$\frac{D_{ik}}{\hat p_k} r_{ik} = D_{ik} r_{ik} + \Big(\frac{D_{ik}}{\hat p_k} - D_{ik}\Big) r_{ik}, \quad \frac{D_{i0}}{\hat p_0} r_{i0} = D_{i0} r_{i0} + \Big(\frac{D_{i0}}{\hat p_0} - D_{i0}\Big) r_{i0}.$$

Summing over $i \in \mathcal{I}_j$ and using (5), the *unweighted* residual sums $\sum D_{ik} r_{ik}$ and $\sum D_{i0} r_{i0}$ vanish *exactly* on each estimation fold, leaving

$$\sum_{i\in\mathcal{I}_j} \hat\psi_{ik} = \sum_{i\in\mathcal{I}_j} \big[\hat m_k(W_i) - \hat m_0(W_i)\big] + \sum_{i\in\mathcal{I}_j} \Big(\tfrac{D_{ik}}{\hat p_k} - D_{ik}\Big) r_{ik} - \sum_{i\in\mathcal{I}_j} \Big(\tfrac{D_{i0}}{\hat p_0} - D_{i0}\Big) r_{i0}.$$

If (as is standard) $\hat p_\ell$ are estimated consistently (e.g., multinomial logit, RF, GBM) on the complementary folds, then the last two sums are $o_p(n)$ by cross-fit orthogonality. Hence the *trimmed* sample moment equation $\sum_i T_i(\hat\tau)\, \hat\psi_{ik} = 0$ is equivalent, up to $o_p(n)$, to the interacted-ATE estimating equation

$$\sum_i T_i(\hat\tau)\, \big[\hat m_k(W_i) - \hat m_0(W_i)\big] = \Big(\sum_i T_i(\hat\tau)\Big)\, \hat\theta_k^{\mathrm{IA}}.$$

Thus the IA estimator is the plug-in DML solution for our score, with $o_p(n^{-1/2})$ differences in $\hat{\theta}_k$ under the ML rates. IA solves the same orthogonal moment as our estimator, modulo cross-fit $o_p(n^{-1/2})$ terms coming from weighting the residuals by $\hat{p}_\ell^{-1}$.

Proposition 5.1 (IA as plug-in DML)**.** *Under Assumption 2.6 (first-stage L_2 rates $o_p(n^{-1/4})$ on the trimmed support) and J-fold cross-fitting, the plug-in DML* sample moment *based on* (4) *equals the IA estimating equation up to $o_p(n^{-1/2})$. Consequently, the IA point estimate and the plug-in DML estimate are asymptotically equivalent:*

$$\hat{\theta}_k^{\mathrm{IA}} - \hat{\theta}_k^{\mathrm{DML}} \;=\; o_p(n^{-1/2}).$$

5.2 Common–Weight (CW): explicit mapping in sample

Let $g(W)$ be the CW weight

$$g(W) = \Big[\sum_{\ell=0}^{K} \hat{\pi}_\ell (1-\hat{\pi}_\ell)/\hat{p}_\ell(W)\Big]^{-1}, \qquad \hat{\pi}_\ell = n^{-1}\sum_i D_{i\ell}.$$

Define the CW "stabilized" indicators

$$\tilde{D}_{i\ell} = \frac{g(W_i)\, D_{i\ell}}{\hat{p}_\ell(W_i)}.$$

The CW normal equation for θ_k is the weighted moment

$$\sum_i T_i(\hat{\tau})\, g(W_i)\, (D_{ik} - \hat{\pi}_k)\, \big(Y_i - \hat{\theta}_k^{\mathrm{CW}}\big) = 0. \tag{6}$$

Start from the raw DML moment $\hat{\psi}_{ik}$ above and multiply the two IPW residual terms by $g(W_i)$ (which does not change the root of the moment condition because the $m_k - m_0$ piece is unweighted). Then

$$\sum_i T_i(\hat{\tau})\, \hat{\psi}_{ik} = \sum_i T_i(\hat{\tau})\big[\hat{m}_k(W_i) - \hat{m}_0(W_i)\big] + \sum_i T_i(\hat{\tau})\tilde{D}_{ik}\, r_{ik} - \sum_i T_i(\hat{\tau})\tilde{D}_{i0}\, r_{i0}.$$

Use $r_{i\ell} = Y_i - \hat{m}_\ell(W_i)$, expand, and regroup terms to isolate Y_i:

$$\sum_i T_i(\hat{\tau})\, \hat{\psi}_{ik} = \sum_i T_i(\hat{\tau})\big[\tilde{D}_{ik} - \tilde{D}_{i0}\big] Y_i \;-\; \sum_i T_i(\hat{\tau})\big[\tilde{D}_{ik} - \tilde{D}_{i0}\big]\hat{m}_\bullet(W_i),$$

where $\hat{m}_\bullet$ denotes the appropriate arm-specific regression in each term. Because the CW propensity fits satisfy the calibration identities

$$\sum_i T_i(\hat{\tau})\, \tilde{D}_{i\ell} = \sum_i T_i(\hat{\tau})\, g(W_i)\, \frac{D_{i\ell}}{\hat{p}_\ell(W_i)} \;\approx\; \sum_i T_i(\hat{\tau})\, g(W_i) \quad (\ell = 0, k),$$

the terms involving $\hat{m}_\bullet$ cancel up to $o_p(n)$, and the remaining Y-part reduces to (6) after subtracting $\hat{\theta}_k \sum_i T_i g(W_i)(D_{ik} - \hat{\pi}_k) = 0$. Thus CW and the plug-in DML moment coincide up to $o_p(n)$ under the CW propensity fit. With CW propensities, the unified score's sample moment is the CW weighted normal equation (plus $o_p(n)$ cross-fit remainders), so the estimators are asymptotically equivalent under the stated rates.

Proposition 5.2 (CW as plug-in DML). *Under Assumption 2.6 and J-fold cross-fitting, the plug-in DML* sample moment *based on* (4) *equals the CW estimating equation up to $o_p(n^{-1/2})$ when CW propensities are used in the score. Hence the CW and plug-in DML point estimates are asymptotically equivalent:*

$$\hat{\theta}_k^{\mathrm{CW}} - \hat{\theta}_k^{\mathrm{DML}} \;=\; o_p(n^{-1/2}).$$

5.3 Easiest-to-Estimate (EW): pairwise AIPW and the cost of discarding data

Restrict to the binary subsample $\{D = 0\} \cup \{D = k\}$ and denote the *binary* propensities by $q_\ell(W) = P(D = \ell \mid W,\, D \in \{0, k\})$. Our score (4) collapses to the classical AIPW score:

$$\psi_{ik}^{\mathrm{AIPW}} = \big[m_k(W_i) - m_0(W_i)\big] + \frac{D_{ik}}{q_k(W_i)}\{Y_i - m_k(W_i)\} - \frac{D_{i0}}{q_0(W_i)}\{Y_i - m_0(W_i)\}.$$

Hence **EW is numerically identical to pairwise AIPW** when implemented with cross-fitted nuisances on the $\{0, k\}$ subsample.

Remark 5.1 (EW uses fewer augmentation samples). *EW throws away all units with $D \notin \{0, k\}$ when forming the sample moment. Those units would still contribute through the augmentation term $m_k(W) - m_0(W)$ in our full-sample score. Heuristically, if $\pi_{0k} = P(D \in \{0, k\})$, then the variance contribution from the augmentation term scales like π_{0k}^{-1} times the conditional variance of $m_k(W) - m_0(W)$ given $D \in \{0, k\}$, so when π_{0k} is small (many-arm designs), EW can be materially less precise. In contrast, the full-sample score averages the augmentation term over* all *W draws, improving precision while keeping the IPW part restricted to the two relevant arms.*

Aggregators and equivalences. Let $T_i(\tau)$ be the trimming indicator (discrete or continuous), and define

$$\hat{\theta}_k(\tau) \;=\; \frac{\sum_{i=1}^n T_i(\tau)\,\varphi_k^\star(Z_i;\hat{\eta})}{\sum_{i=1}^n T_i(\tau)}, \qquad k = 1,\ldots,K,$$

as our cross-fitted orthogonal estimator on the trimmed sample, where $\varphi_k^\star$ is the efficient score and $\hat{\eta}$ collects first-stage learners (as defined earlier in this section).

We consider three scalar aggregators formed as linear functionals of $(\hat{\theta}_1(\tau),\ldots,\hat{\theta}_K(\tau))$:

$$\begin{aligned}
\hat{\theta}^{\mathrm{ATE(IA)}}(\tau) &:= \frac{1}{K}\sum_{k=1}^K \hat{\theta}_k(\tau),\\
\hat{\theta}^{\mathrm{CW}}(\tau) &:= \sum_{k=1}^K \hat{\omega}_k^{\mathrm{CW}}(\tau)\,\hat{\theta}_k(\tau), \quad \text{with } \hat{\omega}_k^{\mathrm{CW}}(\tau) \in \Big\{\frac{1}{K}, \frac{\sum_i T_i(\tau)\mathbf{1}\{D_i = k\}}{\sum_i T_i(\tau)}\Big\},\\
\hat{\theta}^{\mathrm{EW}}(\tau) &:= \sum_{k=1}^K \hat{\omega}_k^{\mathrm{EW}}(\tau)\,\hat{\theta}_k^{\mathrm{pair}}(\tau), \quad \hat{\theta}_k^{\mathrm{pair}}(\tau) := \frac{\sum_i T_i(\tau)\mathbf{1}\{D_i \in \{0,k\}\}\,\varphi_{k,0}^\star(Z_i;\hat{\eta})}{\sum_i T_i(\tau)\mathbf{1}\{D_i \in \{0,k\}\}},
\end{aligned}$$

where $\varphi_{k,0}^\star$ is the standard binary AIPW score for the pair "k vs. 0". The weights $\hat{\omega}_k^{\mathrm{EW}}(\tau)$ can be uniform $(1/K)$ or proportional to trimmed arm shares, mirroring common implementations.

Proposition 5.3 (ATE(IA) equality up to $o_p(n^{-1/2})$)**.** *Suppose cross-fitting and a uniform LLN hold over the trimming grid. Then the interacted-regression ATE computed on the trimmed sample, $\hat{\theta}_{\mathrm{reg}}^{\mathrm{ATE(IA)}}(\tau)$, satisfies*

$$\hat{\theta}_{\mathrm{reg}}^{\mathrm{ATE(IA)}}(\tau) \;=\; \hat{\theta}^{\mathrm{ATE(IA)}}(\tau) \;+\; o_p(n^{-1/2}).$$

Proposition 5.4 (CW equality up to $o_p(n^{-1/2})$)**.** *Let $\hat{\theta}_{\mathrm{reg}}^{\mathrm{CW}}(\tau)$ denote the common-weight regression estimator on the trimmed sample with the same choice of weights $\hat{\omega}^{\mathrm{CW}}(\tau)$. Under the conditions of Proposition 5.3,*

$$\hat{\theta}_{\mathrm{reg}}^{\mathrm{CW}}(\tau) \;=\; \hat{\theta}^{\mathrm{CW}}(\tau) \;+\; o_p(n^{-1/2}).$$

Proposition 5.5 (EW equals pairwise AIPW aggregator)**.** *Let $\hat{\theta}_{\mathrm{1t}}^{\mathrm{EW}}(\tau)$ be the "one-treatment-at-a-time" EW implementation on the trimmed sample. Then*

$$\hat{\theta}_{\mathrm{1t}}^{\mathrm{EW}}(\tau) \;=\; \hat{\theta}^{\mathrm{EW}}(\tau) \;+\; o_p(n^{-1/2}).$$

Moreover, using only $\{D \in \{0,k\}\}$ in $\hat{\theta}_k^{\mathrm{pair}}(\tau)$ is asymptotically less efficient than stacking *all arms in $\hat{\theta}_k(\tau)$ when π_{0k} is small.*

Remark 5.2 (Practical use)**.** *Pick the aggregator (ATE/CW/EW) to match the empirical estimand you wish to report; our theory and standard errors apply componentwise to $(\hat{\theta}_1, \ldots, \hat{\theta}_K)$ and, by the delta method, to any linear aggregator thereof.*

6 Estimation & Inference

6.1 Asymptotic Guarantees

This section answers three questions: *What parameter do we estimate after trimming? How should the trimming threshold $\hat{\tau}$ be chosen? What is the distribution of the data–driven estimator $\hat{\theta}$?*

The proofs rely only on the orthogonality lemmas from Section 3 and the cross-fit regularity results collected in Appendix A.6. A high-level road-map precedes each formal statement; all algebraic details are deferred to the appendix.

Write $T_i(\tau)$ for the trimming indicator, meaning $T_i(\tau) = T_i^{\text{disc}}(\tau)$ in the discrete design and $T_i(\tau) = T_i^{\text{cont}}(\tau)$ in the continuous design. Where no ambiguity can arise we suppress the superscripts and write $T_i(\tau)$.

Explicit indicators:

$$T_i^{\text{disc}}(\tau) = \mathbf{1}\left\{\min_{\ell \le K} p_\ell(W_i) \ge \tau\right\}, \qquad T_i^{\text{cont}}(\tau) = \mathbf{1}\{f(D_i \mid W_i) \ge \tau\}\,.$$

$$\theta_k(\tau) = E\big[\tau_k(W) \,\big|\, T_i^{\text{disc}}(\tau) = 1\big] \;=\; \frac{E\big[\tau_k(W)\, T_i^{\text{disc}}(\tau)\big]}{P\big(T_i^{\text{disc}}(\tau) = 1\big)}, \tag{7}$$

$$\theta^{(c)}(\tau) = E\big[\partial_d m(D_i, W_i) \,\big|\, T_i^{\text{cont}}(\tau) = 1\big] \;=\; \frac{E\big[\partial_d m(D_i, W_i)\, T_i^{\text{cont}}(\tau)\big]}{P\big(T_i^{\text{cont}}(\tau) = 1\big)}. \tag{8}$$

When $\tau = 0$ we recover the full-population parameters θ_k and θ. Because weak overlap forces us to delete the worst-supported units, the random estimator $\hat{\theta}$ targets the corresponding $\theta(\hat{\tau})$. The data-driven trimming bias is controlled below.

Lemma 6.1 (Trimming bias bound)**.** *Let $A_\tau = \{T_i(\tau) = 1\}$ in the discrete case and $A_\tau = \{T_i^{cont}(\tau) = 1\}$ in the continuous case. If either* (i) *$|\tau_k(W)| \le C$ and $|\partial_d m(D, W)| \le C$ a.s.,* *or* (ii) $E[|\tau_k(W)|^q] + E[|\partial_d m(D,W)|^q] \le C$ *for some* $q > 1$, *then as* $\tau \downarrow 0$,

$$|\theta(\tau) - \theta| \;\le\; \begin{cases} C\,P(A_\tau^c), & \textit{under (i)}, \\ C\,P(A_\tau^c)^{\,1-\frac{1}{q}}, & \textit{under (ii)}. \end{cases}$$

Under Assumption 2.4, $P(A_\tau^c) = O(\tau^\delta)$*, so* $|\theta(\tau) - \theta| = O(\tau^\delta)$ *under (i) and* $O(\tau^{\delta(1-1/q)})$ *under (ii).*

Sketch. Write $\theta(\tau) - \theta = \{E[H\mathbf{1}\{A_\tau\}]/P(A_\tau)\} - E[H]$ with $H = \tau_k(W)$ (discrete) or $H = \partial_d m(D,W)$ (continuous); rearrange to $-\{E[(H - EH)\mathbf{1}\{A_\tau^c\}]\}/P(A_\tau)$ and apply boundedness (i) or Hölder (ii). Since $P(A_\tau) \to 1$, the denominators are bounded away from zero for small τ.

6.2 Uniform MSE rule for the threshold

Trimming trades variance reduction against bias. On a finite grid $\mathcal{G} = \{\tau_1 < \cdots < \tau_G\}$ we minimise the empirical proxy of (Bennett et al., 2023) :

$$\widehat{\mathrm{MSE}}(\tau) = \widehat{V}(\tau) + \widehat{B}(\tau)^2, \quad \hat{\tau} = \arg\min_{\tau\in\mathcal{G}} \widehat{\mathrm{MSE}}(\tau).$$

On a finite grid $\mathcal{G}_n$, for each $\tau \in \mathcal{G}_n$ we compute

$$\widehat{\mathrm{MSE}}(\tau) = \underbrace{\widehat{V}(\tau)}_{\text{variance proxy}} + \underbrace{\widehat{B}(\tau)^2}_{\text{bias proxy}}, \tag{9}$$

where (with $\hat{\varphi}_i(\tau)$ the cross-fitted empirical influence value and $\bar{\varphi}_T(\tau) = \{\sum_i T_i(\tau)\}^{-1} \sum_i T_i(\tau)\hat{\varphi}_i(\tau)$)

$$\widehat{V}(\tau) = \Big(\sum_{i=1}^n T_i(\tau)\Big)^{-2} \sum_{i=1}^n T_i(\tau)\left\{\hat{\varphi}_i(\tau) - \bar{\varphi}_T(\tau)\right\}^2, \tag{10}$$

$$\widehat{B}(\tau) = \left|\frac{\sum_{i=1}^n \{1 - T_i(\tau)\}\,\hat{\varphi}_i(\tau)}{\sum_{i=1}^n T_i(\tau)}\right|. \tag{11}$$

The bias proxy (11) exploits that the trimming bias equals $-\{E[\varphi(Z)\mathbf{1}\{T(\tau) = 0\}]\}/P(T(\tau) = 1)$, so (11) is its cross-fitted plug-in. The selector is $\hat{\tau} \in \arg\min_{\tau\in\mathcal{G}_n} \widehat{\mathrm{MSE}}(\tau)$.

Lemma 6.2 (Uniform consistency of the MSE selector)**.** *Under Assumptions 2.1–2.6, 2.4, and 2.5, and the uniform LLN of Lemma A.3, if* $\mathcal{G}_n$ *is finite and* $\max \mathcal{G}_n \downarrow 0$*, then*

$$\sup_{\tau\in\mathcal{G}_n} \big|\widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau)\big| \;\xrightarrow{p}\; 0, \qquad \hat{\tau} \;\xrightarrow{p}\; \tau^\star \in \arg\min_{\tau\in\mathcal{G}_n} \mathrm{MSE}(\tau).$$

Sketch. *Uniform LLNs for* (10) *and* (11) *follow from Lemma A.3 (finite class over $\mathcal{G}_n$), orthogonality (Lemmas 3.1–3.2), and cross-fit stability (Lemma A.4).*

Remark 6.1 (Implementation note)**.** *The replication code uses exactly* (9)–(11) *on the same grid $\mathcal{G}_n$ reported in the paper.*

Lemma A.6 shows $\hat{\tau} \xrightarrow{p} \tau^\star$, the population-optimal threshold, and Lemma A.3 establishes a uniform LLN needed for the subsequent CLT. When clusters are present, the grid search and $\widehat{\mathrm{MSE}}(\tau)$ are computed observation-wise exactly as in the i.i.d. case, but all asymptotics are taken in G (the number of clusters). The uniform LLN over $\mathcal{G}_n$ holds at the cluster level by independence across clusters and the moment bounds in Assumption 6.1.

6.3 Root-n inference with data-driven trimming

Condition 6.1 (Vanishing trimming grid)**.** *Let $\{\mathcal{G}_n\}_{n\geq 1}$ be a sequence of finite trimming grids with $\bar{\tau}_n := \max \mathcal{G}_n \downarrow 0$ and $n\,\bar{\tau}_n^{2\delta} \to 0$ (with δ from Assumption 2.4).*

Remark 6.2. *Under Condition 6.1, the trimming bias is $o_p(n^{-1/2})$ and the centered plug-in variance estimator in Theorem 6.1 is consistent for $V = E[\varphi(Z)^2]$.*

Theorem 6.1 (Central limit theorem)**.** *Assume* (i) *unconfoundedness (Ass. 2.1 and Ass. 2.2),* (ii) *weak overlap (Ass. 2.4),* (iii) *$n^{-1/4}$ first-stage rates (Ass. 2.6), and* (iv) *the vanishing-grid condition (Con. 6.1). Then*

$$\sqrt{n}\,(\hat{\theta} - \theta) \xrightarrow{d} N\big(0, V\big), \qquad V \;=\; E\big[\varphi(Z)^2\big].$$

Moreover (a) the centered plug-in variance estimator

$$\hat{V} \;=\; \Big(\sum_i T_i(\hat{\tau})\Big)^{-2} \sum_{i=1}^{n} T_i(\hat{\tau})\,\Big\{\hat{\varphi}_i - \bar{\varphi}_T\Big\}^2, \qquad \bar{\varphi}_T \;=\; \Big(\sum_i T_i(\hat{\tau})\Big)^{-1} \sum_i T_i(\hat{\tau})\hat{\varphi}_i,$$

is consistent for V, and (b) the trimming bias is $o_p(n^{-1/2})$ because $\sqrt{n}\,|\theta(\hat{\tau}) - \theta| = O_p\big(\sqrt{n}\,\bar{\tau}_n^{\delta}\big) = o_p(1)$.

Remark 6.3 (If trimming does not vanish)**.** *If $\hat{\tau} \xrightarrow{p} \bar{\tau} > 0$ (e.g., a fixed grid not shrinking with n), the estimator targets the trimmed parameter $\theta(\bar{\tau})$ and the ratio form*

$$\hat{\theta} - \theta(\bar{\tau}) \;=\; \frac{\frac{1}{n}\sum_i T_i(\bar{\tau})\,\varphi_i}{\frac{1}{n}\sum_i T_i(\bar{\tau})} \;+\; o_p(n^{-1/2})$$

yields the asymptotic variance

$$V_{\mathrm{trim}}(\bar{\tau}) \;=\; \frac{E\Big[T(\bar{\tau})\left\{\varphi(Z)-(\theta(\bar{\tau})-\theta)\right\}^2\Big]}{\left\{P\big(T(\bar{\tau})=1\big)\right\}^2}.$$

In this regime, the centered plug-in estimator in Theorem 6.1 with $T_i(\hat{\tau})$ consistently estimates $V_{\mathrm{trim}}(\bar{\tau})$. When $\bar{\tau}=0$, $P(T=1)\to 1$ and $\theta(\bar{\tau})-\theta\to 0$, so $V_{\mathrm{trim}}(0)=E[\varphi(Z)^2]=V$.

Sketch of proof. Expand $\sqrt{n}(\hat{\theta}-\theta)$ via the orthogonal score, show that the cross-fit remainder is $o_p(1)$ (Lemma A.4), and control the bias term with Assumption 2.4. A Lyapunov CLT then yields the stated limit; full details are in Appendix A.7. □

Remark 6.4 (Efficiency)**.** *For continuous doses $\varphi^{(c)}$ in* (3) *is the efficient score of Hahn (1998); for discrete arms $\varphi_k^\star$ in* (4) *is likewise efficient. Under the vanishing-grid condition (Condition 6.1), the asymptotic variance equals $V=E[\varphi(Z)^2]$ and data-driven trimming does* not *inflate the variance. If trimming converges to a positive limit, the variance equals the trimmed limit $V_{\mathrm{trim}}(\bar{\tau})$ stated in the remark following Theorem 6.1.*

Corollary 6.2 (Many-arm extension)**.** *If $K=K_n=o(n^{1/3})$ and the score satisfies uniform fourth-moment and variance bounds, Theorem 6.1 holds jointly for $(\hat{\theta}_1,\dots,\hat{\theta}_{K_n})^\top$. Proof is in A.8.*

Even under weak overlap and high-dimensional nuisance learning, the cross-fitted, trimmed DML estimator is root-n regular, efficient, and ready for Wald-type inference.

Remark 6.5 (Why $K_n=o(n^{1/3})$)**.** *Our joint CLT uses a Lyapunov fourth-moment argument for arm-specific influence averages, which yields $K_n=o(n^{1/3})$. High-dimensional Gaussian approximations could in principle relax this (e.g., towards $o(n^{1/2})$ up to logs) under stronger tail regularity and anti-concentration, but we leave that extension for future work.*

6.4 Clustered Sampling: Cross-Fit-by-Cluster and Cluster-Robust Inference

We now formalize inference when observations are grouped into clusters $\{\mathcal{C}_g\}_{g=1}^G$ (e.g., schools, counties, firms) with arbitrary dependence within clusters and independence across clusters.

Assumption 6.1 (Cluster sampling)**.** *The sample consists of $G\to\infty$ clusters $\mathcal{C}_g=\{(Y_{gi},D_{gi},W_{gi})\}_{i=1}^{n_g}$. Clusters are independent and identically distributed draws of arrays of arbitrary size $n_g\ge 1$; dependence within a given cluster is unrestricted. Let $T_{gi}(\tau)$ be the trimming indicator and*

ψ_{gi} the raw moment from Algorithm 1 (either discrete or continuous design). Define the cluster sums

$$\Psi_g(\tau) = \sum_{i \in \mathcal{C}_g} T_{gi}(\tau)\, \psi_{gi}, \qquad N_g(\tau) = \sum_{i \in \mathcal{C}_g} T_{gi}(\tau).$$

Assume $E\big[N_g(\tau)\big] \in (0, \infty)$ and $E\big[\{\Psi_g(\tau) - \theta(\tau)N_g(\tau)\}^2\big] < \infty$ for all τ in the trimming grid $\mathcal{G}_n$ of Assumption 6.1. A uniform fourth-moment bound holds: $\sup_{\tau \in \mathcal{G}_n} E\big[\{\Psi_g(\tau) - \theta(\tau)N_g(\tau)\}^4\big] < \infty$.

Assumption 6.2 (Cross-fitting respects clusters)**.** *Folds are created at the cluster level: each fold j is a subset of clusters $\mathcal{I}_j \subset \{1, \ldots, G\}$, and all observations in $\bigcup_{g \in \mathcal{I}_j} \mathcal{C}_g$ use nuisances trained on the complement $\bigcup_{m \neq j} \bigcup_{g \in \mathcal{I}_m} \mathcal{C}_g$. Assumption 2.6 holds on the trimmed support uniformly over folds.*

Assumption 6.3 (Weak overlap by cluster)**.** *Assumption 2.4 holds and the vanishing-grid condition in Assumption 6.1 is satisfied. In particular, the data-driven $\hat{\tau} \in \mathcal{G}_n$ obeys $\bar{\tau}_n \downarrow 0$ and $n\, \bar{\tau}_n^{2\delta} \to 0$, where $n = \sum_{g=1}^{G} n_g$.*

Theorem 6.3 (Cluster-robust CLT)**.** *Under Assumptions 6.1–6.3 and the orthogonality lemmas in Section 3, let*

$$\hat{\theta}(\hat{\tau}) = \frac{\sum_{g=1}^{G} \Psi_g(\hat{\tau})}{\sum_{g=1}^{G} N_g(\hat{\tau})}.$$

Then

$$\sqrt{G}\, \big\{\hat{\theta}(\hat{\tau}) - \theta\big\} \xrightarrow{d} \mathcal{N}\Big(0,\ V_{\text{clu}}\Big), \quad V_{\text{clu}} = \frac{\big(\Psi_g - \theta N_g\big)}{\{E[N_g]\}^2},$$

where all moments are evaluated at the population-optimal limiting threshold (zero under Condition 6.1). Moreover, the following plug-in cluster-robust variance estimator is consistent:

$$\hat{V}_{\text{clu}} = \Big(\sum_{g=1}^{G} N_g(\hat{\tau})\Big)^{-2} \frac{G}{G-1} \sum_{g=1}^{G} \Big\{\hat{\Phi}_g(\hat{\tau})\Big\}^2, \qquad \hat{\Phi}_g(\hat{\tau}) = \sum_{i \in \mathcal{C}_g} T_{gi}(\hat{\tau})\, \big\{\hat{\psi}_{gi} - \hat{\theta}(\hat{\tau})\big\}.$$

(Notice $\sum_g \hat{\Phi}_g(\hat{\tau}) = 0$ by construction, so no extra centering is needed.) Using t-critical values with $G-1$ degrees of freedom is recommended when G is small.

Proof sketch. Write $\bar{\Psi}_G = G^{-1} \sum_g \Psi_g(\hat{\tau})$ and $\bar{N}_G = G^{-1} \sum_g N_g(\hat{\tau})$; then $\hat{\theta}(\hat{\tau}) = \bar{\Psi}_G / \bar{N}_G$. A delta-method expansion in the i.i.d. cluster array yields $\sqrt{G}\{\hat{\theta} - \theta\} = \{E[N_g]\}^{-1} G^{-1/2} \sum_g \{\Psi_g - \theta N_g\} + o_p(1)$. Orthogonality and cross-fit-by-cluster give $o_p(1)$ remainders at the same rate as in Theorem 6.1, while the trimming bias is $o_p(G^{-1/2})$ because $\bar{\tau}_n \downarrow 0$ and $G \to \infty$. Lyapunov's CLT applies to the cluster sums $\Phi_g = \Psi_g - \theta N_g$ under the fourth-moment bound.

Consistency of $\hat{V}_{\text{clu}}$ follows from a cluster-level LLN and the fact that $\sum_g \hat{\Phi}_g = 0$. Full details in A.14 mirror the non-clustered proof with indices aggregated at the cluster level. □

Remark 6.6 (What changes in practice?)**.** *(i) Create folds at the cluster level; (ii) keep the* point estimator *unchanged; (iii) compute standard errors from cluster sums* $\hat{\Phi}_g(\hat{\tau})$ *as in Theorem 6.3; (iv) use* t_{G-1} *critical values if* G *is small.*

7 Monte-Carlo Simulations

7.1 Data Generation Process

Before turning to empirical results we describe the data-generating processes (DGPs) that will be used to evaluate the estimators introduced in the preceding sections. Two families of DGPs are considered.

7.1.1 Design A: three discrete arms

Generate ten baseline covariates

$$W = (W_1, \ldots, W_{10}) \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1).$$

The arm indicators D_{ik} follow a multinomial logit:

$$p_k(W) = \frac{\exp\{\gamma_k + \gamma\, W^\top \beta_k\}}{1 + \sum_{\ell=1}^{3} \exp\{\gamma_\ell + \gamma\, W^\top \beta_\ell\}}, \quad k = 1, 2, 3,$$

with intercepts $\gamma_1 = \gamma_2 = \gamma_3 = 0$ and sparse loadings $\beta_{k1} = 1,\ \beta_{k2} = -1,\ \beta_{kj} = 0\ (j \geq 3)$. The single scalar γ controls *overlap*: $\gamma = 0.6$ (*strong*), $\gamma = 1.8$ (*weak*).

Write

$$Y = \mu(W) + \sum_{k=1}^{3} D_{ik}\, \tau_k(W) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

with

$$\mu(W) = W_1 + W_2^2, \quad \begin{cases} \tau_k^{\text{lin}}(W) = \delta_k + \delta\, W_1, \\ \tau_k^{\text{nonlin}}(W) = \delta_k + \delta\, \sin(W_1 + W_2), \end{cases} \quad (\delta_1, \delta_2, \delta_3) = (1, 2, 3),\ \delta = 0.5.$$

Design 7.1.1 crosses $\{\text{strong}, \text{weak}\} \times \{\text{linear}, \text{non-linear}\}$ to give four discrete scenarios.

7.1.2 Design B: continuous dosage

Draw the dose D from a truncated normal on $[-1, 1]$:

$$D \sim \mathcal{TN}_{[-1,1]}(\mu_d(W),\, \sigma_d^2), \quad \mu_d(W) = 0.4\,W_1, \;\; \sigma_d^2 = 0.25.$$

The conditional density $f(d \mid W)$ is therefore proportional to $\exp\bigl[-(d-\mu_d(W))^2/(2\sigma_d^2)\bigr]$.

$$m(d, W) = 1 + \sin\bigl[(d+1)\,W_2\bigr] + 0.3\,d^2, \quad Y = m(D, W) + \varepsilon, \varepsilon \sim \mathcal{N}(0, 1).$$

Weak-overlap variants use trimming indicators $\mathbf{1}\{f(D \mid W) \geq \tau_n\}$ with $\tau_n \in \{0.02,\, 0.01\}$.

7.1.3 Learners and estimators

Nuisance learners (common to all estimators).

Propensities: gradient-boosted trees (100 trees, depth 2, learning rate 0.05; `lightgbm`).

Outcome regressions: honest causal forests (2 000 trees; `grf::causal_forest`). Conditional density (Design 7.1.2 only): quadratic local-polynomial log-density.

Estimators compared.

(a) Naïve linear/partial-linear regression.

(b) Pairwise AIPW baseline on $\{D \in \{0, k\}\}$ with cross-fitted nuisances (numerically identical to EW under our implementation).

(c) GPHK's EW, IA, and CW estimators (Design 7.1.1 only).

(d) Proposed orthogonal DML: $\hat{\theta}_k^{\text{DML}}$ for arms and $\hat{\theta}^{\text{DML,C}}$ for dose.

We estimate $\log f(d \mid W)$ by a boundary-corrected local-polynomial regression of order 2 with product kernels; the derivative $\partial_d \log f$ is obtained analytically from the local fit. Bandwidths are selected by 4-fold cross-validation minimizing the negative conditional log-likelihood on held-out folds. A sensitivity table varying the bandwidth multiplier in $\{0.5, 1, 2\}$ is reported in Appendix **??**.

All methods use $J = 4$-fold cross-fitting. Trimming thresholds are selected on the grid $\{0.05,\, 0.02,\, 0.01,\, 0.005\}$ via the MSE criterion of Kallus and Oprescu (2023). The next section presents bias, variance, and RMSE results.

Table 1: Monte-Carlo data-generation settings

Feature	**A – Linear**	**A – Non-linear**	**B – Continuous**
Covariates W		$\mathcal{N}(0,1)^{\otimes 10}$	
Overlap lever	$\gamma = 0.6$ / 1.8	$\gamma = 0.6$ / 1.8	$\tau_n = 0.02$ / 0.01
Treatment effect shape	$\delta_k + \delta W_1$	$\delta_k + \delta \sin(W_1 + W_2)$	Quadratic + trigonometric
Baseline mean $\mu(W)$	$W_1 + W_2^2$	$W_1 + W_2^2$	—
Sample sizes n		1 000 2 000 4 000	
Replications		1 000	

Notes: "AIPW(0,k)" coincides numerically with EW (pairwise AIPW on the $\{0, k\}$ subsample with cross-fitted nuisances). We therefore omit a separate AIPW column to avoid redundancy; full outputs are in the replication package.

7.2 Evidence from Simulations

Where you summarize contributions, add a clause: ... A data-driven trimming rule handles weak overlap; and, under independent clusters, a cross-fit-by-cluster scheme with a cluster-summed variance estimator yields valid cluster-robust inference (Theorem 6.3).

This section documents the finite–sample performance of the orthogonal DML scores and the main competitors under the four variants of Design A (three discrete treatment levels) and under the continuous–dose Design B. All programmes are contained in the `dmlContam` replication package. Every experiment is replicated $R = 1{,}000$ times with four–fold cross–fitting; without sample–splitting the orthogonal scores inherit own–observation bias (see Appendix A.6 for cross-fit regularity). For each design we report Monte–Carlo bias, root mean–squared error (RMSE) and simulation standard error at $n \in \{1{,}000, 2{,}000, 4{,}000\}$.

The left panel of Table 2 (strong overlap, $\gamma = 0.6$) shows that the orthogonal DML scores are virtually unbiased in every size–arm cell; the maximal absolute bias is 0.08 at $n = 1{,}000$ and shrinks with n, becoming negligible by $n = 4{,}000$ (consistent with the $o_p(n^{-1/2})$ remainder of cross-fitted DML). The fully–interacted OLS benchmark (IA) shares this trait because its outcome model is correctly specified. EW is numerically identical to pairwise AIPW on the $\{0, k\}$ subsample with cross-fitted nuisances, so it is unbiased under our implementation. Its remaining drawback is efficiency: discarding augmentation samples increases variance relative to the full-sample orthogonal score (Remark 5.1). Small nonzero Monte-Carlo means reported in the tables reflect finite-sample noise rather than systematic "flattening." The substantive distinction that remains is efficiency: EW discards all $D \notin \{0, k\}$ units from the augmentation term, which increases variance relative to the full-sample orthogonal score

Table 2: Finite–sample *linear* performance (Design A)

(a) Strong overlap ($\gamma = 0.6$)

n	Metric	Estimator				
				IA	CW	OLS
Panel A: Arm 1 ($\tau_1 = 1$)						
	Bias	0.074	−0.065	0.074	−1.024	0.488
1 000	RMSE	0.150	0.150	0.150	1.021	0.520
	s.e.	0.004	0.004	0.004	0.000	0.006
	Bias	0.046	−0.073	0.046	−1.030	0.488
2 000	RMSE	0.099	0.110	0.100	1.026	0.514
	s.e.	0.003	0.003	0.003	0.000	0.004
	Bias	0.034	−0.071	0.034	−1.033	0.490
4 000	RMSE	0.073	0.090	0.073	1.029	0.502
	s.e.	0.002	0.002	0.002	0.000	0.003
Panel B: Arm 2 ($\tau_2 = 2$)						
	Bias	0.070	−0.067	0.070	−1.976	0.484
1 000	RMSE	0.144	0.152	0.144	1.982	0.522
	s.e.	0.004	0.004	0.004	0.000	0.006
	Bias	0.053	−0.066	0.053	−1.972	0.490
2 000	RMSE	0.101	0.113	0.103	1.968	0.510
	s.e.	0.003	0.003	0.003	0.000	0.004
	Bias	0.031	−0.075	0.031	−1.969	0.488
4 000	RMSE	0.071	0.096	0.072	1.966	0.503
	s.e.	0.002	0.002	0.002	0.000	0.003
Panel C: Arm 3 ($\tau_3 = 3$)						
	Bias	0.074	−0.063	0.074	−2.929	0.483
1 000	RMSE	0.135	0.142	0.135	2.928	0.520
	s.e.	0.004	0.004	0.004	0.000	0.006
	Bias	0.047	−0.073	0.047	−2.914	0.488
2 000	RMSE	0.101	0.112	0.101	2.908	0.512
	s.e.	0.003	0.003	0.003	0.000	0.004
	Bias	0.035	−0.070	0.035	−2.905	0.490
4 000	RMSE	0.072	0.090	0.072	2.905	0.501
	s.e.	0.002	0.002	0.002	0.000	0.003

(b) Weak overlap ($\gamma = 1.8$)

n	Metric	Estimator				
				IA	CW	OLS
Panel A: Arm 1 ($\tau_1 = 1$)						
	Bias	0.263	−0.116	0.263	−1.014	0.926
1 000	RMSE	0.310	0.210	0.310	1.010	0.940
	s.e.	0.005	0.006	0.005	0.000	0.006
	Bias	0.186	−0.171	0.186	−1.018	0.917
2 000	RMSE	0.230	0.230	0.230	1.020	0.930
	s.e.	0.004	0.005	0.004	0.000	0.004
	Bias	0.147	−0.179	0.147	−1.020	0.920
4 000	RMSE	0.170	0.210	0.170	1.020	0.920
	s.e.	0.003	0.003	0.003	0.000	0.003
Panel B: Arm 2 ($\tau_2 = 2$)						
	Bias	0.252	−0.128	0.252	−1.986	0.917
1 000	RMSE	0.300	0.210	0.300	1.990	0.930
	s.e.	0.005	0.005	0.005	0.000	0.006
	Bias	0.187	−0.171	0.187	−1.983	0.920
2 000	RMSE	0.230	0.220	0.230	1.980	0.930
	s.e.	0.004	0.004	0.004	0.000	0.004
	Bias	0.145	−0.182	0.145	−1.982	0.917
4 000	RMSE	0.170	0.210	0.170	1.980	0.920
	s.e.	0.003	0.003	0.003	0.000	0.003
Panel C: Arm 3 ($\tau_3 = 3$)						
	Bias	0.252	−0.128	0.252	−2.958	0.915
1 000	RMSE	0.300	0.220	0.300	2.960	0.930
	s.e.	0.005	0.006	0.005	0.000	0.006
	Bias	0.186	−0.171	0.186	−2.949	0.918
2 000	RMSE	0.230	0.220	0.230	2.950	0.930
	s.e.	0.004	0.004	0.004	0.000	0.004
	Bias	0.145	−0.181	0.145	−2.945	0.915
4 000	RMSE	0.170	0.210	0.170	2.940	0.920
	s.e.	0.003	0.003	0.003	0.000	0.003

Notes: Bias is the mean deviation from the true ATE. Monte–Carlo s.e. equals the simulated sd divided by $\sqrt{1000}$.

(Remark 5.1). For completeness, we include a *pairwise AIPW* baseline (binary AIPW on $\{0, k\}$); by construction it is numerically identical to EW in our implementation, so we report a single column for both. Contamination weighting (CW) inherits the largest negative bias (between -1.0 and -2.9), while conventional OLS propagates upward–sloping bias across arms.

RMSE mirrors these rankings. DML and IA fall from 0.15 to 0.07 as n quadruples, tracking the parametric $n^{-1/2}$ rate. EW stagnates around $0.09-0.11$; CW and OLS are at least one order of magnitude less precise. Consequently, DML realises RMSE gains exceeding 70% relative to OLS in every cell and dominates EW once $n \geq 4{,}000$.

Weakening the overlap to $\gamma = 1.8$ (right panel of Table 2) removes 14% of the sample at $n = 1{,}000$ and 8% at $n = 4{,}000$. Even so, DML retains root–n precision and keeps absolute bias below 0.26; its RMSE declines from 0.31 to 0.17, overtaking EW (whose RMSE rises to 0.21). CW and OLS remain dominated, with order–one bias and RMSE. The adaptive mean–squared–error rule of Section 6.2 selects thresholds between 1% and 2% in 99% of replications, so the trimming–induced ATE bias is $o_p(n^{-1/2})$.

Table 3 reports the same estimators when the true CATE is sinusoidal. Under strong overlap DML remains nearly unbiased (maximal absolute bias 0.08), but EW now enjoys a clear efficiency edge: its RMSE falls from 0.13 to 0.06, whereas DML and IA settle around 0.08. Once overlap is weak the qualitative ordering tightens. DML cuts its RMSE by one third as the sample grows, yet EW improves even faster and preserves a pronounced efficiency lead. CW is still plagued by severe negative bias (-1.0 to -3.0), and OLS reproduces the contamination pattern observed in the linear design.

Table 3: Finite–sample *non-linear* performance (Design A)

(a) Strong overlap ($\gamma = 0.6$)

n	Metric	DML	EW	IA	CW	OLS
		Estimator				
Panel A: Arm 1 ($\tau_1 = 1$)						
	Bias	0.083	0.016	0.083	−1.023	0.414
1 000	RMSE	0.150	0.130	0.150	1.020	0.450
	s.e.	0.004	0.004	0.004	0.0001	0.006
	Bias	0.059	0.012	0.059	−1.029	0.414
2 000	RMSE	0.100	0.090	0.100	1.030	0.430
	s.e.	0.003	0.003	0.003	0.0001	0.004
	Bias	0.046	0.013	0.046	−1.032	0.417
4 000	RMSE	0.080	0.060	0.080	1.030	0.430
	s.e.	0.002	0.002	0.002	0.0001	0.003
Panel B: Arm 2 ($\tau_2 = 2$)						
	Bias	0.078	0.011	0.078	−1.976	0.409
1 000	RMSE	0.150	0.130	0.150	1.980	0.450
	s.e.	0.004	0.004	0.004	0.0001	0.006
	Bias	0.065	0.019	0.065	−1.970	0.417
2 000	RMSE	0.110	0.090	0.110	1.970	0.440
	s.e.	0.003	0.003	0.003	0.0001	0.004
	Bias	0.044	0.010	0.044	−1.968	0.414
4 000	RMSE	0.070	0.060	0.070	1.970	0.420
	s.e.	0.002	0.002	0.002	0.0001	0.003
Panel C: Arm 3 ($\tau_3 = 3$)						
	Bias	0.083	0.017	0.083	−2.928	0.411
1 000	RMSE	0.150	0.130	0.150	2.930	0.450
	s.e.	0.004	0.004	0.004	0.0002	0.006
	Bias	0.059	0.012	0.059	−2.913	0.414
2 000	RMSE	0.100	0.090	0.100	2.910	0.430
	s.e.	0.003	0.003	0.003	0.0001	0.004
	Bias	0.046	0.014	0.046	−2.904	0.416
4 000	RMSE	0.070	0.060	0.070	2.900	0.430
	s.e.	0.002	0.002	0.002	0.0001	0.003

(b) Weak overlap ($\gamma = 1.8$)

n	Metric	DML	EW	IA	CW	OLS
		Estimator				
Panel A: Arm 1 ($\tau_1 = 1$)						
	Bias	0.236	−0.035	0.236	−1.014	0.754
1 000	RMSE	0.290	0.200	0.290	1.010	0.770
	s.e.	0.005	0.006	0.005	0.0001	0.005
	Bias	0.183	−0.050	0.183	−1.017	0.747
2 000	RMSE	0.230	0.180	0.230	1.020	0.760
	s.e.	0.005	0.005	0.005	0.0001	0.004
	Bias	0.159	−0.032	0.159	−1.019	0.750
4 000	RMSE	0.190	0.130	0.190	1.020	0.750
	s.e.	0.003	0.004	0.003	0.0001	0.003
Panel B: Arm 2 ($\tau_2 = 2$)						
	Bias	0.224	−0.049	0.224	−1.986	0.746
1 000	RMSE	0.280	0.200	0.280	1.990	0.770
	s.e.	0.005	0.006	0.005	0.0001	0.005
	Bias	0.184	−0.052	0.184	−1.983	0.749
2 000	RMSE	0.230	0.170	0.230	1.980	0.760
	s.e.	0.004	0.005	0.004	0.0001	0.004
	Bias	0.157	−0.035	0.157	−1.981	0.746
4 000	RMSE	0.190	0.130	0.190	1.980	0.750
	s.e.	0.003	0.004	0.003	0.0001	0.003
Panel C: Arm 3 ($\tau_3 = 3$)						
	Bias	0.226	−0.047	0.226	−2.957	0.745
1 000	RMSE	0.290	0.210	0.290	2.960	0.760
	s.e.	0.006	0.006	0.006	0.0001	0.005
	Bias	0.184	−0.049	0.184	−2.949	0.747
2 000	RMSE	0.230	0.170	0.230	2.950	0.760
	s.e.	0.004	0.005	0.004	0.0001	0.004
	Bias	0.158	−0.035	0.158	−2.944	0.744
4 000	RMSE	0.190	0.130	0.190	2.940	0.750
	s.e.	0.003	0.004	0.003	0.0001	0.003

Notes: Bias is the mean deviation from the true ATE. Monte–Carlo s.e. equals the simulated sd divided by $\sqrt{R}$ with $R = 1{,}000$ replications.

		Estimator					
n	Metric	DML	EW	IA	CW	OLS	OWDML
Panel A: Arm 1 ($\tau_1 = 1$)							
	Bias	0.083	0.016	0.125	−1.023	0.414	−0.949
1,000	RMSE	0.150	0.130	0.214	1.023	0.451	0.949
	s.e.	0.004	0.004	0.005	0.000	0.006	0.000
	Bias	0.059	0.012	0.123	−1.029	0.414	−0.946
2,000	RMSE	0.102	0.086	0.175	1.029	0.434	0.946
	s.e.	0.003	0.003	0.004	0.000	0.004	0.000
	Bias	0.046	0.013	0.125	−1.032	0.417	−0.944
4,000	RMSE	0.075	0.061	0.152	1.032	0.426	0.944
	s.e.	0.002	0.002	0.003	0.000	0.003	0.000
Panel B: Arm 2 ($\tau_2 = 2$)							
	Bias	0.078	0.011	0.119	−1.976	0.409	−1.902
1,000	RMSE	0.148	0.130	0.206	1.976	0.446	1.902
	s.e.	0.004	0.004	0.005	0.000	0.006	0.000
	Bias	0.065	0.019	0.128	−1.970	0.417	−1.894
2,000	RMSE	0.107	0.089	0.179	1.970	0.437	1.894
	s.e.	0.003	0.003	0.004	0.000	0.004	0.000
	Bias	0.044	0.010	0.121	−1.968	0.414	−1.890
4,000	RMSE	0.073	0.059	0.149	1.968	0.423	1.890
	s.e.	0.002	0.002	0.003	0.000	0.003	0.000
Panel C: Arm 3 ($\tau_3 = 3$)							
	Bias	0.083	0.017	0.126	−2.928	0.411	−2.855
1,000	RMSE	0.149	0.129	0.213	2.928	0.447	2.855
	s.e.	0.004	0.004	0.005	0.000	0.006	0.000
	Bias	0.059	0.012	0.123	−2.913	0.414	−2.842
2,000	RMSE	0.104	0.088	0.176	2.913	0.432	2.842
	s.e.	0.003	0.003	0.004	0.000	0.004	0.000
	Bias	0.046	0.014	0.125	−2.904	0.416	−2.836
4,000	RMSE	0.074	0.060	0.152	2.904	0.425	2.836
	s.e.	0.002	0.002	0.003	0.000	0.003	0.000

Notes: Bias is the mean deviation from the true ATE. Monte–Carlo s.e. equals the simulated sd divided by $\sqrt{R}$ with $R = 1\,000$ replications.

n	Metric	**Estimator** DML	EW	IA	CW	OLS	OWDML
Panel A: Arm 1 ($\tau_1 = 1$)							
	Bias	0.236	−0.035	0.306	−1.014	0.754	−0.960
1,000	RMSE	0.293	0.200	0.362	1.014	0.774	0.960
	s.e.	0.005	0.006	0.006	0.000	0.005	0.000
	Bias	0.183	−0.050	0.307	−1.017	0.747	−0.959
2,000	RMSE	0.234	0.180	0.338	1.017	0.757	0.959
	s.e.	0.005	0.005	0.004	0.000	0.004	0.000
	Bias	0.159	−0.032	0.308	−1.019	0.750	−0.959
4,000	RMSE	0.190	0.130	0.323	1.019	0.754	0.959
	s.e.	0.003	0.004	0.003	0.000	0.003	0.000
Panel B: Arm 2 ($\tau_2 = 2$)							
	Bias	0.224	−0.049	0.302	−1.986	0.746	−1.928
1,000	RMSE	0.283	0.199	0.361	1.986	0.766	1.928
	s.e.	0.005	0.006	0.006	0.000	0.005	0.000
	Bias	0.184	−0.052	0.303	−1.983	0.749	−1.923
2,000	RMSE	0.229	0.170	0.332	1.983	0.758	1.923
	s.e.	0.004	0.005	0.004	0.000	0.004	0.000
	Bias	0.157	−0.035	0.304	−1.981	0.746	−1.922
4,000	RMSE	0.190	0.135	0.319	1.981	0.751	1.922
	s.e.	0.003	0.004	0.003	0.000	0.003	0.000
Panel C: Arm 3 ($\tau_3 = 3$)							
	Bias	0.226	−0.047	0.301	−2.957	0.745	−2.896
1,000	RMSE	0.285	0.206	0.360	2.957	0.764	2.896
	s.e.	0.006	0.006	0.006	0.000	0.005	0.000
	Bias	0.184	−0.049	0.307	−2.949	0.747	−2.888
2,000	RMSE	0.230	0.170	0.336	2.949	0.757	2.888
	s.e.	0.004	0.005	0.004	0.000	0.004	0.000
	Bias	0.158	−0.035	0.300	−2.944	0.744	−2.886
4,000	RMSE	0.189	0.132	0.315	2.944	0.750	2.886
	s.e.	0.003	0.004	0.003	0.000	0.003	0.000

Notes: Bias is the mean deviation from the true ATE. Monte–Carlo s.e. equals the simulated sd divided by $\sqrt{R}$ with $R = 1\,000$ replications.

(i) Orthogonal DML simultaneously eliminates contamination bias and achieves root–n accuracy; CW and OLS achieve neither, while EW is only single–robust.

(ii) In linear environments IA shadows DML, but when the outcome equation is misspecified

Table 4: Design B (continuous dose): Monte–Carlo summary

n	Metric	DML	OLS	TRUE
		Estimator		
1,000	Bias	−0.001	−0.002	0.000
	RMSE	0.085	0.084	0.023
	s.e.	0.003	0.003	0.001
2,000	Bias	0.001	−0.001	0.000
	RMSE	0.061	0.061	0.017
	s.e.	0.002	0.002	0.001
4,000	Bias	0.001	0.000	0.000
	RMSE	0.044	0.044	0.012
	s.e.	0.001	0.001	0.000

Notes: Bias is the mean deviation from the true APE. Monte–Carlo s.e. equals the simulated sd divided by $\sqrt{R}$ with R replications.

or overlap is weak the DML efficiency premium reaches 40%−50% in RMSE.

(iii) EW is competitive only under generous overlap and near–linear response surfaces; its own–observation bias and slower convergence render it inferior in weak–overlap or highly non–linear settings.

Results for the continuous treatment are collected in Table **??**. Trimming thresholds $\tau_n \in \{0.02, 0.01\}$ correspond to the 1.6-th and 2.9-th empirical density percentiles at the smallest sample size. Naïve partial–linear regression is grossly contaminated: its bias exceeds 0.9 at $n = 1{,}000$ and *increases* with n as the kernel regressors extrapolate into low–density regions. The proposed continuous–dose DML score removes 98% of that bias and beats OLS in RMSE by a factor three. Precision hardly suffers from trimming: the RMSE gap between $\tau = 0.02$ and the infeasible $\tau = 0$ is below 4%.

(a) Contamination bias can exceed sampling variance even at $n = 4{,}000$; its removal is empirically consequential.

(b) Orthogonal DML preserves $\sqrt{n}$ accuracy in *all* scenarios, whereas fixes that lack double robustness (EW) or algebraic debiasing (CW, OLS) deteriorate outside their design points.

(c) In continuous doses, replacing a partial–linear regression with the proposed estimator yields order–of–magnitude bias reductions and RMSE gains between 50% and 70%.

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8 Conclusion

This paper develops a single, Neyman-orthogonal influence-function framework that *algebraically purges* the contamination bias recently highlighted by Goldsmith-Pinkham *et al.* (2024). In both multi-arm and continuous-dose settings, our cross-fitted DML estimators achieve $\sqrt{n}$ consistency and semiparametric efficiency even when (i) nuisance functions are high-dimensional and non-linear, and (ii) overlap is weak enough to require trimming. The theory nests GPHK's three "formal fixes" as special cases, extends to a growing number of arms, and holds under data-driven trimming that minimises an estimated mean-squared error. Monte-Carlo evidence shows up to 50 percent RMSE gains relative to existing approaches; two empirical illustrations confirm that practical effect magnitudes can shift substantially once contamination is removed. An open-source package, `dmlContam`, implements the entire workflow and facilitates adoption.

Three caveats warrant emphasis. First, our weak-overlap results require $n\tau_n^{2\delta} \to 0$ and thus do not cover scenarios in which overlap deteriorates faster than $n^{-1/(2\delta)}$; relaxing this rate—or replacing trimming by overlap-adaptive weights—is a key topic for future research. Second, we allow a growing number of arms and provide a joint CLT when $K_n = o(n^{1/3})$; all fixed-arm results place no restriction on K_n. Third, We establish cluster-robust inference under independent clusters (Theorem 6.3). Extending the framework to dependence structures beyond independent clusters (e.g., two-way clustering or serial correlation in panels) remains an open direction.

Beyond addressing these limitations, we see three natural extensions: (i) cluster-robust variants that accommodate group-level experimental designs; (ii) dynamic treatments or dosage paths, where contamination can propagate through time; and (iii) causal discovery pipelines in which the proposed EIFs act as drop-in scoring rules for model-selection heuristics. More broadly, our findings underscore the value of orthogonal machine-learning estimators: they can deliver classical root-n precision while freeing empirical researchers from linearity and strict overlap assumptions that are rarely satisfied in practice.

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A Proofs of Main and Auxiliary Results

A.1 Notation and Additional Symbols

Throughout the proofs $Z_i = (Y_i, D_i, W_i)$ collects the outcome, treatment, and covariates for observation i. The treatment takes one of $K+1$ mutually–exclusive values, $D_i \in \{0, 1, \ldots, K\}$, and we write $D_{ik} = 1\{D_i = k\}$ for the arm–k dummy. Propensity scores are denoted $p_k(W) = P(D_i = k \mid W_i = W)$. For the baseline (control) arm the conditional mean outcome is $\mu(W) = E[Y_i \mid D_i = 0, W_i = W]$.

It is useful to centre the treatment indicators by defining $X_{ik} = D_{ik} - p_k(W_i)$; these satisfy $E[X_{ik} \mid W] = 0$ and $E[X_{ik}^2 \mid W] = p_k(1 - p_k)$. The conditional average treatment effect for arm k is $\tau_k(W) = E[\,Y_i(k) - Y_i(0) \mid W_i = W]$.

Weak overlap is handled by trimming observations whose propensity scores are too close to zero. The trimming indicator is therefore $T_i = \mathbf{1}\{\min_k p_k(W_i) \geq \tau_n\}$ in the discrete–arm setting; for a continuous treatment an analogous rule based on the conditional density is used.

Finally, φ_i always denotes the population influence function—either (3) for continuous doses or (4) for discrete arms—evaluated at the true nuisance functions.

A.2 Proof of Lemma 3.1

Recall the score function for the continuous treatment case is

$$\varphi^{(c)}(Z; \theta, \eta) = s(D, W)\{Y - m(D, W)\} + \partial_d m(D, W) - \theta,$$

where the nuisance parameter is $\eta = (m, f)$ and the score is $s(d, W) = \partial_d \log f(d|W) = (\partial_d f(d|W))/f(d|W)$.

(i) Identification. We show that $E[\varphi^{(c)}(Z; \theta, \eta)] = 0$ if and only if θ is the true average partial effect. By the law of iterated expectations,

$$\begin{aligned} E[\varphi^{(c)}(Z; \theta, \eta)] &= E\big[s(D, W)\{Y - m(D, W)\}\big] + E\big[\partial_d m(D, W)\big] - \theta \\ &= E_W\left[E_{D|W}\big[s(D, W)\{Y - m(D, W)\}\big|W\big]\right] + E\big[\partial_d m(D, W)\big] - \theta. \end{aligned}$$

The inner expectation is taken over D conditional on W:

$$E_{D|W}\big[s(D, W)\{Y - m(D, W)\}\big|W\big] = \int_{\mathcal{D}} s(d, W)\left(E[Y|D = d, W] - m(d, W)\right) f(d|W) dd.$$

At the true nuisance function $m(d, W) = E[Y|D = d, W]$ (by Assumption 2.2), the term in parentheses is identically zero for all d. Thus, the first term vanishes: $E\big[s(D, W)\{Y - m(D, W)\}\big] = 0$. The moment condition then simplifies to:

$$E[\varphi^{(c)}(Z; \theta, \eta)] = E\big[\partial_d m(D, W)\big] - \theta.$$

This expectation is zero if and only if $\theta = E[\partial_d m(D, W)]$, which confirms that the score correctly identifies the target parameter.

(ii) Neyman Orthogonality (Pathwise Derivative). We now show that the estimator is insensitive to small, regular perturbations in the nuisance functions. Let $\eta_r = (m + rh_m, f + rh_f)$ be a perturbed nuisance parameter along a path indexed by r, where $h = (h_m, h_f)$ is a regular perturbation in the nuisance tangent space. The score along this path is $s_r(d, W) = (\partial_d f_r(d|W))/f_r(d|W)$. The moment condition as a function of r is

$$K(r) = E\left[s_r(D, W)\{Y - m_r(D, W)\} + \partial_d m_r(D, W) - \theta\right].$$

We seek to show that the Gateaux derivative $\left.\frac{dK(r)}{dr}\right|_{r=0}$ is zero. We compute the derivative term by term.

1. Derivative of the first component: Using the product rule,

$$\begin{aligned} \left.\frac{d}{dr} E\left[s_r\{Y - m_r\}\right]\right|_{r=0} &= E\left[\left(\left.\frac{ds_r}{dr}\right|_{r=0}\right)\{Y - m\} + s\left(\left.\frac{d(Y - m_r)}{dr}\right|_{r=0}\right)\right] \\ &= E\left[\left(\left.\frac{ds_r}{dr}\right|_{r=0}\right)\{Y - m\} - s(D, W)h_m(D, W)\right]. \end{aligned}$$

The term $E[\cdot \times \{Y - m\}]$ is an expectation of a random variable multiplied by the true population residual $Y - m(D,W)$. By the law of iterated expectations, this term is zero because $E[Y - m(D,W)|D,W] = 0$. Thus, this part of the derivative simplifies to $-E[s(D,W)h_m(D,W)]$.

2. Derivative of the second component:

$$\frac{d}{dr}E\left[\partial_d m_r(D,W)\right]\bigg|_{r=0} = E\left[\partial_d h_m(D,W)\right].$$

Combining the terms, the full Gateaux derivative is:

$$\left.\frac{dK(r)}{dr}\right|_{r=0} = E\left[\partial_d h_m(D,W) - s(D,W)h_m(D,W)\right].$$

We now show this expression equals zero by using integration by parts.

$$\begin{aligned}
E\left[\partial_d h_m - s h_m\right] &= E_W\left[\int_{\mathcal{D}} \left(\partial_d h_m(d,W)\right) f(d|W)dd - \int_{\mathcal{D}} s(d,W)h_m(d,W)f(d|W)dd\right] \\
&= E_W\left[\int_{\mathcal{D}} \left(\partial_d h_m(d,W)\right) f(d|W)dd - \int_{\mathcal{D}} \frac{\partial_d f(d|W)}{f(d|W)}h_m(d,W)f(d|W)dd\right] \\
&= E_W\left[\int_{\mathcal{D}} \left(f(d|W)\partial_d h_m(d,W) - h_m(d,W)\partial_d f(d|W)\right)dd\right].
\end{aligned}$$

The integrand is the result of the product rule for derivatives: $f \cdot \partial_d h_m + h_m \cdot \partial_d f = \partial_d(f \cdot h_m)$. Therefore, the expression becomes:

$$E_W\left[\int_{\mathcal{D}} \partial_d\left(f(d|W)h_m(d,W)\right)dd\right].$$

By the Fundamental Theorem of Calculus, this integral evaluates to the function at the boundaries of the support $\mathcal{D}$:

$$E_W\left[\left[f(d|W)h_m(d|W)\right]_{d=\inf\mathcal{D}}^{d=\sup\mathcal{D}}\right].$$

By our explicit Boundary Condition (Assumption 3.1), this term is zero. Thus, the Gateaux derivative is zero:

$$\left.\frac{dK(r)}{dr}\right|_{r=0} = 0.$$

This confirms that the moment condition is Neyman-orthogonal to the nuisance parameter $\eta = (m, f)$, completing the proof. □

A.3 Auxiliary identity for the multi-arm score

For $k \in \{1, \ldots, K\}$ write

$$X_{ik} \;:=\; \frac{D_{ik} - p_k(W_i)}{1 - p_k(W_i)}, \qquad p_k(W) = P(D_i = k \mid W_i),$$

and recall that $m_k(W) = E[Y_i \mid D_i = k, W_i]$ as well as $\tau_k(W) = m_k(W) - m_0(W)$ for $k \geq 1$.

Lemma A.1. *For every square-integrable $\xi(W)$ and each arm k the following two identities hold:*

$$(a) \quad E\big[X_{ik}\, \xi(W_i)\big] = 0, \tag{12}$$

$$(b) \quad E\big[X_{ik}\, \{Y_i - m_k(W_i)\} \mid W_i\big] = p_k(W_i)\, \tau_k(W_i). \tag{13}$$

Proof. (a) By construction $E[D_{ik} \mid W_i] = p_k(W_i)$; hence

$$E\big[X_{ik} \mid W_i\big] = \frac{E[D_{ik} \mid W_i] - p_k(W_i)}{1 - p_k(W_i)} = 0,$$

and taking the unconditional expectation yields (12).

(b) Write the potential–outcome decomposition

$$Y_i \;=\; \mu(W_i) + \sum_{\ell=1}^{K} D_{i\ell}\, \tau_\ell(W_i) + \varepsilon_i, \qquad E[\varepsilon_i \mid D_i, W_i] = 0,$$

with $\mu(W) = m_0(W)$. Because $m_k(W) = \mu(W) + \tau_k(W)$ we have

$$Y_i - m_k(W_i) \;=\; \sum_{\ell=1}^{K} D_{i\ell}\, \tau_\ell(W_i) - \tau_k(W_i) + \varepsilon_i.$$

Multiplying by X_{ik} and taking conditional expectation,

$$E\big[X_{ik}\, \{Y_i - m_k(W_i)\} \mid W_i\big] \;=\; p_k(W_i)\, \tau_k(W_i),$$

because $E[X_{ik} \mid W_i] = 0$ by part *(a)* of the lemma and $E[\varepsilon_i \mid D_i, W_i] = 0$. This establishes (13). $\square$

A.4 Proof of Lemma 3.2

Recall the influence function for the ATE of arm k versus control:

$$\varphi_k^\star(Z;\theta_k,\eta) = \underbrace{m_k(W) - m_0(W)}_{A_1} + \underbrace{\frac{D_{ik}}{p_k(W)}(Y - m_k(W))}_{A_2} - \underbrace{\frac{D_{i0}}{p_0(W)}(Y - m_0(W))}_{A_3} - \theta_k.$$

The nuisance parameter is $\eta = (m_0, \dots, m_K, p_0, \dots, p_K)$.

(i) Identification and Double Robustness. We first show that $E[\varphi_k^\star] = 0$ if either (a) the outcome models m_ℓ are correct or (b) the propensity models p_ℓ are correct. Let $E[Y|D = \ell, W] = m_\ell^*(W)$ and $P(D = \ell|W) = p_\ell^*(W)$ be the true conditional functions.

Case 1: Outcome models are correct ($m_\ell = m_\ell^*$). By the law of iterated expectations and unconfoundedness (Assumptions 2.1 and 2.2):

$$\begin{aligned}
E[A_2] &= E\left[\frac{D_{ik}}{p_k(W)}(Y - m_k(W))\right] = E_W\left[E\left[\frac{D_{ik}}{p_k(W)}(Y - m_k(W))\middle| W\right]\right] \\
&= E_W\left[\frac{1}{p_k(W)}P(D_i = k|W)E[Y - m_k(W)|D_i = k, W]\right] \\
&= E_W\left[\frac{p_k^*(W)}{p_k(W)}(m_k^*(W) - m_k(W))\right] = 0.
\end{aligned}$$

Similarly, $E[A_3] = 0$. The total expectation is $E[\varphi_k^\star] = E[m_k^*(W) - m_0^*(W)] - \theta_k$. This is zero if $\theta_k = E[Y(k) - Y(0)]$.

Case 2: Propensity models are correct ($p_\ell = p_\ell^*$).

$$\begin{aligned}
E[A_2] &= E\left[\frac{D_{ik}}{p_k^*(W)}\{Y - m_k(W)\}\right] = E_W\left[\frac{1}{p_k^*(W)}E[D_{ik}Y - D_{ik}m_k(W)|W]\right] \\
&= E_W\left[\frac{1}{p_k^*(W)}(p_k^*(W)m_k^*(W) - p_k^*(W)m_k(W))\right] = E[m_k^*(W) - m_k(W)].
\end{aligned}$$

Similarly, $E[A_3] = E[m_0^*(W) - m_0(W)]$. The total expectation is:

$$\begin{aligned}
E[\varphi_k^\star] &= E[m_k(W) - m_0(W)] + E[m_k^*(W) - m_k(W)] - E[m_0^*(W) - m_0(W)] - \theta_k \\
&= E[m_k^*(W) - m_0^*(W)] - \theta_k.
\end{aligned}$$

This is zero if $\theta_k = E[Y(k) - Y(0)]$. In either case, the score correctly identifies the ATE.

(ii) Neyman Orthogonality. We show that the moment is locally insensitive to perturbations in η. Let $\eta_r = \eta_0 + rh$ be a perturbation, where η_0 is the true nuisance parameter and $h = (h_m, h_p)$ is a regular perturbation. We compute the Gateaux derivative of

$K(r) = E[\varphi_k^\star(Z;\theta_k,\eta_r)]$ at $r = 0$. The derivative $\frac{dK(r)}{dr}\Big|_{r=0}$ is the expectation of the sum of derivatives of $\varphi_k^\star$ with respect to each nuisance function, multiplied by the corresponding perturbation.

- Perturbation of m_k by $h_{m,k}$: The terms in $\varphi_k^\star$ depending on m_k are $m_k(W) - \frac{D_{ik}}{p_k(W)} m_k(W)$. The derivative w.r.t. m_k is $\left(1 - \frac{D_{ik}}{p_k(W)}\right)$. The contribution to the Gateaux derivative is $E\left[\left(1 - \frac{D_{ik}}{p_k(W)}\right) h_{m,k}(W)\right]$. By iterated expectations, $E\left[1 - \frac{D_{ik}}{p_k(W)} \middle| W\right] = 1 - \frac{p_k(W)}{p_k(W)} = 0$. So this term is zero.

- Perturbation of m_0 by $h_{m,0}$: The terms are $-m_0(W) + \frac{D_{i0}}{p_0(W)} m_0(W)$. The derivative w.r.t. m_0 is $\left(\frac{D_{i0}}{p_0(W)} - 1\right)$. The contribution is $E\left[\left(\frac{D_{i0}}{p_0(W)} - 1\right) h_{m,0}(W)\right]$, which is zero by the same logic.

- Perturbation of p_k by $h_{p,k}$: The term is $\frac{D_{ik}}{p_k(W)}\{Y - m_k(W)\}$. The derivative w.r.t. p_k is $-\frac{D_{ik}}{p_k(W)^2}\{Y - m_k(W)\}$. The contribution is $E\left[-\frac{D_{ik}}{p_k(W)^2}\{Y - m_k(W)\} h_{p,k}(W)\right]$. This expectation is zero because we evaluate at the true nuisance η_0, where $m_k(W) = m_k^*(W)$. As shown in the identification part, $E[D_{ik}\{Y - m_k^*(W)\}|W] = 0$. Thus, this term is zero.

- Perturbation of p_0 by $h_{p,0}$: The term is $-\frac{D_{i0}}{p_0(W)}\{Y - m_0(W)\}$. The derivative w.r.t. p_0 is $+\frac{D_{i0}}{p_0(W)^2}\{Y - m_0(W)\}$. The contribution $E\left[\frac{D_{i0}}{p_0(W)^2}\{Y - m_0(W)\} h_{p,0}(W)\right]$ is zero by the same logic.

- Perturbations of m_ℓ, p_ℓ for $\ell \notin \{0, k\}$: The score $\varphi_k^\star$ does not depend on these nuisance functions, so the derivatives are zero.

Since all components of the Gateaux derivative are zero, we have $\frac{d}{dr} E[\varphi_k^\star(Z;\theta_k,\eta_r)]\big|_{r=0} = 0$. This establishes Neyman orthogonality. □

A.5 Proof of Theorem 3.1

Throughout this proof the trimming threshold τ_n is *fixed by design* (no data–dependent tuning) and satisfies $n\tau_n^{2\delta} \to 0$ with the δ in Assumption 2.4. Write

$$T_i = \mathbf{1}\{\min_{k \le K_n} p_k(W_i) \ge \tau_n\}, \qquad \varphi_i = \varphi(Z_i;\theta_k,\eta_0),$$

and let $\hat{\eta}_i = \hat{\eta}^{(-j(i))}$ be the fold-specific nuisance vector used for observation i.

1. Moment expansion. The DML estimator solves $\frac{1}{n}\sum_{i=1}^{n} T_i\, \psi(Z_i; \hat{\theta}_k, \hat{\eta}_i) = 0$, and $\partial_\theta \psi \equiv -1$. A first-order Taylor expansion around θ_k gives

$$\sqrt{n}\,(\hat{\theta}_k - \theta_k) \;=\; \frac{1}{\sqrt{n}}\sum_{i=1}^{n} T_i\, \psi(Z_i; \theta_k, \hat{\eta}_i) + o_p(1). \tag{14}$$

Add and subtract the *population* influence function to obtain

$$\sqrt{n}(\hat{\theta}_k - \theta_k) \;=\; \underbrace{\frac{1}{\sqrt{n}}\sum_{i=1}^{n} T_i\varphi_i}_{S_n} + \underbrace{R_n}_{\text{remainder}} + \underbrace{B_n}_{\text{bias}} + o_p(1),$$

where

$$R_n = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} T_i\big\{\psi(Z_i; \theta_k, \hat{\eta}_i) - \varphi_i\big\}, \qquad B_n = \sqrt{n}\,\big\{\theta_k(\tau_n) - \theta_k\big\}.$$

2. Bounding the remainder $R_n = o_p(1)$. Decompose $R_n = R_{n,1} + R_{n,2}$ with

$$R_{n,1} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}(T_i - \hat{T}_i)\psi(Z_i; \theta_k, \hat{\eta}_i), \qquad R_{n,2} = \frac{1}{\sqrt{n}}\sum_{i=1}^{n} T_i\big\{\psi(Z_i; \theta_k, \hat{\eta}_i) - \varphi_i\big\}.$$

Indicator error ($R_{n,1}$). By Lemma A.5, uniformly over $\tau \in \mathcal{G}_n$, $\Pr\big(\hat{T}_i(\tau) \neq T_i(\tau)\big) = o(n^{-1/2})$ under Assumptions 2.6 and 2.5. Hence $R_{n,1} = o_p(1)$ uniformly on the grid.

Nuisance error ($R_{n,2}$). Write $\hat{\eta}_i = \eta_0 + \Delta_i$. The mean-value expansion

$$\psi(Z_i; \theta_k, \hat{\eta}_i) - \varphi_i = \partial_\eta \psi(Z_i; \theta_k, \eta_0)[\Delta_i] + \tfrac{1}{2}\,\partial_\eta^2 \psi(Z_i; \theta_k, \tilde{\eta}_i)[\Delta_i, \Delta_i], \quad \tilde{\eta}_i \text{ between } \eta_0 \text{ and } \hat{\eta}_i,$$

shows that every term in $R_{n,2}$ is either (i) linear in one nuisance error or (ii) quadratic in two errors. Because the score is Neyman-orthogonal (Lemmas 3.1–3.2) the *population* expectation of the linear term vanishes, and the empirical average is $o_p(1)$ by the cross-fit stability Lemma A.4. The quadratic remainder is bounded via Cauchy–Schwarz:

$$E\Big[T_i\,|\,\partial_\eta^2 \psi[\Delta_i, \Delta_i]|\Big] \;\lesssim\; \|\Delta_i\|_{L^2}^2 = o_p(n^{-1/2}),$$

so its $\sqrt{n}$-scaled sum is $o_p(1)$. Thus $R_{n,2} = o_p(1)$ and hence $R_n = o_p(1)$.

3. Trimming bias $B_n = o_p(1)$. By Assumption 2.4 $|\theta_k(\tau_n) - \theta_k| \le C\tau_n^\delta$, so $|B_n| = \sqrt{n}\,|\theta_k(\tau_n) - \theta_k| \le C\sqrt{n}\,\tau_n^\delta = o(1)$ because $n\tau_n^{2\delta} \to 0$.

4. Asymptotic normality of S_n. Define $X_{ni} = n^{-1/2}T_i\varphi_i$. The array $\{X_{ni}\}_{i\le n}$ is i.i.d. across i with $E[X_{ni}] = 0$ and $\sum_{i=1}^{n} \mathrm{Var}(X_{ni}) = E[T_i\varphi_i^2] \to V_k$ by Lemma A.2. A Lyapunov (or Lindeberg) condition holds because $E[\varphi_i^4] < \infty$ uniformly on $\{T_i = 1\}$. Hence $S_n =$

$\sum_{i=1}^n X_{ni} \xrightarrow{d} N(0, V_k)$.

5. Putting the pieces together. Gathering terms,

$$\sqrt{n}(\hat{\theta}_k - \theta_k) \;=\; S_n + R_n + B_n + o_p(1) \;=\; S_n + o_p(1) \xrightarrow{d} N(0, V_k).$$

6. Consistency of the plug-in variance estimator. Write $\hat{V}_k = \left(\frac{1}{n}\sum_i T_i\right)^{-2} \frac{1}{n}\sum_{i=1}^n T_i\, \psi(Z_i; \hat{\theta}_k, \hat{\eta}_i)^2$. Because T_i is deterministic given W_i, an ordinary LLN plus Lemma A.4 shows $\frac{1}{n}\sum_i T_i \to_p P(T_i = 1) = 1$ and $\frac{1}{n}\sum_i T_i \psi_i^2 \to_p E[T_i \varphi_i^2] = V_k$. Therefore $\hat{V}_k \xrightarrow{p} V_k$.

$\square$

A.6 Regularity Lemmas for Trimmed Cross–Fitting

Throughout this subsection $T_i(\tau) = \mathbf{1}\{\min_k p_k(W_i) \geq \tau\}$ (or its density analogue) denotes the trimming indicator, and $\varphi_i = \varphi(Z_i; \theta, \eta)$ is the *population* influence function.

Lemma A.2 (Finite moments on the trimmed support)**.** *If Assumptions 2.4–2.6 hold and* $\sup_{k \leq K_n} E[\varphi_k^\star(Z)^4] < \infty$*, then for every fixed* $\tau > 0$

$$\sup_{i \leq n} E\big[T_i(\tau)\, \varphi_i^2\big] < \infty, \qquad \sup_{i \leq n} E\big[T_i(\tau)\, \varphi_i^4\big] < \infty.$$

Proof. On $\{p_k(W) \geq \tau\}$ we have $D_{ik}/p_k(W) \leq 1/\tau$; the definition of φ_i and the bounded fourth moment assumption give the result. $\square$

Lemma A.3 (Uniform LLN on a data–dependent trimmed set)**.** *Let* $\mathcal{G} = \{\tau_1 < \cdots < \tau_G\}$ *be the finite grid in Section 6.2. Then*

$$\sup_{\tau \in \mathcal{G}} \Big| \frac{1}{n}\sum_{i=1}^n T_i(\tau)\varphi_i - E[T_i(\tau)\varphi_i] \Big| = o_p(1), \qquad \sup_{\tau \in \mathcal{G}} \Big| \frac{1}{n}\sum_{i=1}^n T_i(\tau)\varphi_i^2 - E[T_i(\tau)\varphi_i^2] \Big| = o_p(1).$$

Proof. The function classes $\{T(\tau)\varphi \colon \tau \in \mathcal{G}\}$ and $\{T(\tau)\varphi^2 \colon \tau \in \mathcal{G}\}$ are finite; combine Lemma A.2 with the finite-class LLN (Lemma 2.4 of Van der Vaart and Wellner, 1996). $\square$

Lemma A.4 (Cross-fit stability of the empirical score)**.** *Let* $\hat{\eta}^{(-j)}$ *be the fold-*j *nuisance estimate and define* $\hat{\varphi}_i = \varphi(Z_i; \theta, \hat{\eta}^{(-j(i))})$*. Under Assumptions 2.6–2.4,*

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n T_i(\tau)\big\{\hat{\varphi}_i - \varphi_i\big\} = o_p(1) \quad \textit{uniformly in } \tau \in \mathcal{G}.$$

Sketch. Expand $\hat{\varphi}_i - \varphi_i$ by a first-order Gateaux derivative; each term is a product of *one* score component and *one* $n^{-1/4}$-rate first-stage error. Cauchy–Schwarz, the ML rate in

Assumption 2.6, and Lemma A.2 yield a bound of order $n^{-1/4}$; multiplying by $\sqrt{n}$ gives $o_p(1)$. See Appendix B.3 for an identical calculation in the proof of Theorem 3.1. $\square$

Lemmas A.2–A.4 together justify (i) the uniform MSE estimator for τ, (ii) the plug-in variance formula, and (iii) all stochastic equicontinuity steps in the main CLT.

Lemma A.5 (Indicator disagreement under margin)**.** *Under Assumptions 2.6 and 2.5, uniformly over $\tau \in \mathcal{G}_n$,*

$$\Pr\big(\hat{T}_i(\tau) \neq T_i(\tau)\big) \;\leq\; C \sum_{\ell \leq K} E\big[\, |\hat{p}_\ell^{(-i)}(W_i) - p_\ell(W_i)|^\kappa \big] \;=\; o\big(n^{-1/2}\big).$$

An identical bound holds in the continuous case with (p_ℓ) replaced by f.

Proof sketch. Use $\{\hat{T} \neq T\} \subset \{\, |\min_\ell p_\ell(W) - \tau| \leq \Delta_i \,\}$, $\Delta_i = \max_\ell |\hat{p}_\ell - p_\ell|$, the margin Assumption 2.5, and the trimmed L_2 rates from Assumption 2.6, interpolating $\|\cdot\|_2$ to κ-moments on the trimmed support. $\square$

A.7 Proof of Theorem 6.1

For simplicity, we present the proof for a generic parameter θ and score $\psi(Z; \theta, \eta)$ that satisfies the conditions of the theorem. Let η_0 be the true nuisance functions, $\hat{\eta}_i$ be the cross-fitted estimate for observation i, and $\varphi_i = \psi(Z_i; \theta_0, \eta_0)$ be the true influence function. Let $\hat{\tau}$ be the data-driven trimming threshold.

The DML estimator $\hat{\theta}$ is defined as the solution to $\frac{1}{n}\sum_{i=1}^n \hat{T}_i(\hat{\tau})\psi(Z_i; \hat{\theta}, \hat{\eta}_i) = 0$, where $\hat{T}_i(\tau) = \mathbf{1}\{\min_k \hat{p}_k^{(-i)}(W_i) \geq \tau\}$. The proof consists of three main parts: establishing a linear representation for the estimator, showing the asymptotic normality of the leading term, and proving the consistency of the variance estimator.

1. Linear Representation of $\sqrt{n}(\hat{\theta} - \theta_0)$. Since $\partial_\theta \psi \equiv -1$, a first-order Taylor expansion of the moment condition around the true parameter θ_0 yields:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \hat{T}_i(\hat{\tau})\psi(Z_i; \theta_0, \hat{\eta}_i) + o_p(1).$$

We decompose the leading term on the right-hand side by adding and subtracting terms involving the true influence function φ_i and the true trimming indicator $T_i(\tau) = \mathbf{1}\{\min_k p_k(W_i) \geq \tau\}$:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n T_i(\hat{\tau})\varphi_i}_{S_n\text{: Main Term}} + \underbrace{R_n}_{\text{Remainder}} + \underbrace{B_n}_{\text{Bias}} + o_p(1). \tag{15}$$

The remainder R_n and bias B_n are defined as:

$$R_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \left[\hat{T}_i(\hat{\tau})\psi(Z_i; \hat{\eta}_i) - T_i(\hat{\tau})\varphi_i\right]$$
$$B_n = \sqrt{n}E[(T_i(\hat{\tau}) - 1)\varphi_i].$$

We now show that both R_n and B_n are asymptotically negligible.

Bounding the Remainder ($R_n = o_p(1)$). The remainder can be split into two parts: $R_n = R_{n,1} + R_{n,2}$, where

$$R_{n,1} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} (\hat{T}_i(\hat{\tau}) - T_i(\hat{\tau}))\psi(Z_i; \hat{\eta}_i) \qquad \text{(Indicator Error)}$$
$$R_{n,2} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} T_i(\hat{\tau})(\psi(Z_i; \hat{\eta}_i) - \varphi_i) \qquad \text{(Nuisance Error)}$$

The Nuisance Error Term ($R_{n,2}$). To show $R_{n,2} = o_p(1)$, we must explicitly derive the difference $\psi(Z_i; \hat{\eta}_i) - \varphi_i$. We do so for the discrete-arm score φ_k^*; the logic for the continuous case is analogous. Let $\hat{\eta} = (\hat{m}, \hat{p})$ and $\eta_0 = (m, p)$.

$$\begin{aligned}\psi_k^*(Z; \hat{\eta}) - \psi_k^*(Z; \eta_0) = ([\hat{m}_k - m_k] - [\hat{m}_0 - m_0]) \\ + \left(\frac{D_{ik}}{\hat{p}_k}(Y - \hat{m}_k) - \frac{D_{ik}}{p_k}(Y - m_k)\right) \\ - \left(\frac{D_{i0}}{\hat{p}_0}(Y - \hat{m}_0) - \frac{D_{i0}}{p_0}(Y - m_0)\right).\end{aligned}$$

Consider the term for arm k. By adding and subtracting $\frac{D_{ik}}{p_k}(Y - \hat{m}_k)$, we get:

$$\begin{aligned}\frac{D_{ik}}{\hat{p}_k}(Y - \hat{m}_k) - \frac{D_{ik}}{p_k}(Y - m_k) &= \left(\frac{D_{ik}}{\hat{p}_k}(Y - \hat{m}_k) - \frac{D_{ik}}{p_k}(Y - \hat{m}_k)\right) + \left(\frac{D_{ik}}{p_k}(Y - \hat{m}_k) - \frac{D_{ik}}{p_k}(Y - m_k)\right) \\ &= D_{ik}(Y - \hat{m}_k)\left(\frac{1}{\hat{p}_k} - \frac{1}{p_k}\right) - \frac{D_{ik}}{p_k}(\hat{m}_k - m_k) \\ &= -\frac{D_{ik}(Y - \hat{m}_k)}{p_k\hat{p}_k}(\hat{p}_k - p_k) - \frac{D_{ik}}{p_k}(\hat{m}_k - m_k).\end{aligned}$$

Substituting this back into the full expression for the difference and rearranging terms gives:

$$\psi_k^*(\hat{\eta}) - \varphi_k^*(\eta_0) = \underbrace{(\hat{m}_k - m_k)\left(1 - \frac{D_{ik}}{p_k}\right) - (\hat{m}_0 - m_0)\left(1 - \frac{D_{i0}}{p_0}\right)}_{\text{Term (I): Pathwise derivative w.r.t. } m}$$
$$\underbrace{-\frac{D_{ik}(Y - m_k)}{p_k^2}(\hat{p}_k - p_k) + \frac{D_{i0}(Y - m_0)}{p_0^2}(\hat{p}_0 - p_0)}_{\text{Term (II): Pathwise derivative w.r.t. } p}$$
$$\underbrace{-\frac{D_{ik}(Y - m_k)}{p_k}\left(\frac{1}{\hat{p}_k} - \frac{1}{p_k} + \frac{\hat{p}_k - p_k}{p_k^2}\right)(\hat{p}_k - p_k) + \dots}_{\text{Term (III): Second-order and higher terms}}$$
$$\underbrace{+\frac{D_{ik}}{p_k\hat{p}_k}(\hat{m}_k - m_k)(\hat{p}_k - p_k) - \dots}_{\text{Term (IV): Cross-product terms}}$$

Neyman orthogonality implies that the expectation of Terms (I) and (II) is zero. The DML framework with cross-fitting is designed specifically to ensure that the *sample average* of these first-order pathwise derivatives is asymptotically negligible. The remaining terms, (III) and (IV), are products of at least two nuisance function errors. For example, a typical cross-product term is $\frac{D_{ik}}{p_k\hat{p}_k}(\hat{m}_k - m_k)(\hat{p}_k - p_k)$. The contribution of such terms to $\sqrt{n}R_{n,2}$ is bounded by sums like:

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} T_i(\hat{\tau})\frac{D_{ik}}{p_k\hat{p}_k}(\hat{m}_k^{(-i)} - m_k)(\hat{p}_k^{(-i)} - p_k).$$

By the Cauchy-Schwarz inequality, the expectation of the absolute value of this term is bounded by:

$$\sqrt{n} \cdot E\left[\frac{T_i}{p_k\hat{p}_k}\right] \cdot \|\hat{m}_k - m_k\|_{L^2(P)} \cdot \|\hat{p}_k - p_k\|_{L^2(P)}.$$

Under Assumption 2.5, this is $\sqrt{n} \cdot O(1) \cdot o_p(n^{-1/4}) \cdot o_p(n^{-1/4}) = o_p(1)$. A formal argument using empirical process theory confirms that the sum of all such second-order terms is $o_p(1)$. Thus, $R_{n,2} = o_p(1)$, and the full remainder is $R_n = o_p(1)$.

Bounding the Trimming Bias ($B_n = o_p(1)$). The bias term B_n captures the effect of discarding observations with low estimated propensity scores. Its definition is:

$$B_n = \sqrt{n}E[(T_i(\hat{\tau}) - 1)\varphi_i] = -\sqrt{n}E[\mathbf{1}\{T_i(\hat{\tau}) = 0\}\varphi_i].$$

To bound its magnitude, we apply the Cauchy-Schwarz inequality for expectations, which states that for any two random variables X and Y, $|E[XY]| \leq \sqrt{E[X^2]E[Y^2]}$. Let $X = \varphi_i$

and $Y = \mathbf{1}\{T_i(\hat{\tau}) = 0\}$.

$$\begin{aligned}
|B_n| &= \sqrt{n}\,|E[\varphi_i \cdot \mathbf{1}\{T_i(\hat{\tau}) = 0\}]| \\
&\leq \sqrt{n}\sqrt{E[\varphi_i^2]} \cdot \sqrt{E\left[(\mathbf{1}\{T_i(\hat{\tau}) = 0\})^2\right]} && \text{(by Cauchy-Schwarz)} \\
&= \sqrt{n}\sqrt{V} \cdot \sqrt{E[\mathbf{1}\{T_i(\hat{\tau}) = 0\}]} && \text{(since } \mathbf{1}^2 = \mathbf{1}\text{)} \\
&= \sqrt{n}\sqrt{V} \cdot \sqrt{P(T_i(\hat{\tau}) = 0)}.
\end{aligned}$$

By Assumption 2.4, the probability of being trimmed is controlled by the threshold τ_n:

$$P(T_i(\hat{\tau}) = 0) = P(\min_k p_k(W) < \hat{\tau}) = O_p(\hat{\tau}^\delta).$$

Substituting this into our bound gives:

$$|B_n| \leq \sqrt{n} \cdot \sqrt{V} \cdot \sqrt{O_p(\hat{\tau}^\delta)} = O_p(\sqrt{n} \cdot \hat{\tau}^{\delta/2}).$$

The theorem's condition requires $n\tau_n^\delta \to 0$ for all fixed τ_n on the grid. Since Lemma A.6 shows that the data-driven $\hat{\tau}$ converges in probability to a value on this grid (or zero), the condition implies that $n\hat{\tau}^\delta \to_p 0$. This is equivalent to $\sqrt{n}\hat{\tau}^{\delta/2} \to_p 0$. Therefore, the bias is asymptotically negligible: $B_n = o_p(1)$.

2. Asymptotic Normality. The preceding steps establish the linear representation of the estimator:

$$\sqrt{n}(\hat{\theta} - \theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} T_i(\hat{\tau})\varphi_i + o_p(1).$$

Let $S_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n T_i(\hat{\tau})\varphi_i$ and let $X_{ni} = n^{-1/2}T_i(\hat{\tau})\varphi_i$. We verify the conditions for the Lyapunov Central Limit Theorem for the triangular array $\{X_{ni}\}$.

(a) Zero Mean (Asymptotically): The mean of each term in the sum is $E[X_{ni}] = n^{-1/2}E[T_i(\hat{\tau})\varphi_i]$. As shown above, $\sqrt{n}E[T_i(\hat{\tau})\varphi_i] = -B_n = o_p(1)$, so $E[X_{ni}] = o_p(n^{-1})$.

(b) Convergent Variance: The sum of the variances is:

$$\sum_{i=1}^{n} \mathrm{Var}(X_{ni}) = \sum_{i=1}^{n} \frac{1}{n}\mathrm{Var}(T_i(\hat{\tau})\varphi_i) = \mathrm{Var}(T_1(\hat{\tau})\varphi_1) = E[T_1(\hat{\tau})^2\varphi_1^2] - (E[T_1(\hat{\tau})\varphi_1])^2.$$

As $n \to \infty$, $\hat{\tau} \xrightarrow{p} 0$, which implies $T_1(\hat{\tau}) \xrightarrow{p} 1$. By the Dominated Convergence Theorem (since $T_1^2\varphi_1^2 \leq \varphi_1^2$ and $E[\varphi_1^2] < \infty$), the first term converges to $E[\varphi_1^2] = V$. The squared mean term converges to $(E[\varphi_1])^2 = 0$. Thus, $\sum_{i=1}^n \mathrm{Var}(X_{ni}) \to V$.

(c) Lyapunov Condition: We must show that for some $\epsilon > 0$, the sum of higher moments vanishes. Let ϵ be such that $E[|\varphi_i|^{2+\epsilon}] < \infty$ (a finite fourth moment as in Corollary 7.2 is sufficient, implying $\epsilon = 2$).

$$\begin{aligned}\sum_{i=1}^{n} E[|X_{ni}|^{2+\epsilon}] &= \sum_{i=1}^{n} \frac{1}{n^{(2+\epsilon)/2}} E[|T_i(\hat{\tau})\varphi_i|^{2+\epsilon}] \\ &= n \cdot \frac{1}{n^{1+\epsilon/2}} E[|T_i(\hat{\tau})\varphi_i|^{2+\epsilon}] \\ &= n^{-\epsilon/2} E[T_i(\hat{\tau})^{2+\epsilon}|\varphi_i|^{2+\epsilon}] \le n^{-\epsilon/2} E[|\varphi_i|^{2+\epsilon}].\end{aligned}$$

Since $E[|\varphi_i|^{2+\epsilon}]$ is a finite constant, the expression is $O(n^{-\epsilon/2})$, which converges to 0 as $n \to \infty$.

With all conditions satisfied, the Lyapunov CLT implies that $S_n \xrightarrow{d} N(0, V)$. We now apply Slutsky's Theorem, which states that if $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$, then $X_n + Y_n \xrightarrow{d} X + c$. In our case, $X_n = S_n$ and $Y_n = \sqrt{n}(\hat{\theta} - \theta_0) - S_n = R_n - B_n = o_p(1)$. Thus, $Y_n \xrightarrow{p} 0$. Slutsky's Theorem gives:

$$\sqrt{n}(\hat{\theta} - \theta_0) = S_n + o_p(1) \xrightarrow{d} N(0, V) + 0 = N(0, V).$$

3. Consistency of the Variance Estimator. Let $\bar{\varphi}_T = (\sum_i T_i(\hat{\tau}))^{-1} \sum_i T_i(\hat{\tau})\hat{\varphi}_i$. By Lemma A.3 and Assumption 6.1, $\bar{\varphi}_T = o_p(1)$ and

$$\frac{1}{n}\sum_i T_i(\hat{\tau})\big\{\hat{\varphi}_i - \bar{\varphi}_T\big\}^2 \xrightarrow{p} E\big[\varphi(Z)^2\big].$$

Because $\big(\frac{1}{n}\sum_i T_i(\hat{\tau})\big)^{-2} \to_p 1$, the centered plug-in estimator stated in Theorem 6.1 is consistent for V. In the non-vanishing case $\hat{\tau} \to_p \bar{\tau} > 0$, the same argument yields consistency for $V_{\text{trim}}(\bar{\tau})$ (see the remark following Theorem 6.1).

Thus, we only need to show the consistency of the numerator: $\frac{1}{n}\sum_i \hat{T}_i\psi_i(\hat{\theta}, \hat{\eta})^2 \xrightarrow{p} V$. We decompose the error using the triangle inequality:

$$\left|\frac{1}{n}\sum_i \hat{T}_i\hat{\psi}_i^2 - V\right| \le \underbrace{\left|\frac{1}{n}\sum_i \hat{T}_i\hat{\psi}_i^2 - \frac{1}{n}\sum_i T_i\varphi_i^2\right|}_{\text{Term A}} + \underbrace{\left|\frac{1}{n}\sum_i T_i\varphi_i^2 - V\right|}_{\text{Term B}}.$$

- Term B converges to 0: By the Law of Large Numbers, since $T_i(\hat{\tau})\varphi_i^2$ are i.i.d. and

$\hat{\tau} \xrightarrow{p} 0$:

$$\frac{1}{n}\sum_i T_i(\hat{\tau})\varphi_i^2 \xrightarrow{p} E[T_i(0)\varphi_i^2] = E[\varphi_i^2] = V.$$

- Term A converges to 0: We can write $|\hat{T}_i\hat{\psi}_i^2 - T_i\varphi_i^2| \leq |(\hat{T}_i - T_i)\hat{\psi}_i^2| + |T_i(\hat{\psi}_i^2 - \varphi_i^2)|$. The first part vanishes because $P(\hat{T}_i \neq T_i) \to 0$. For the second part, note that $\hat{\psi}_i^2 - \varphi_i^2 = (\hat{\psi}_i - \varphi_i)(\hat{\psi}_i + \varphi_i)$. Since $\hat{\theta} \xrightarrow{p} \theta_0$ and $\|\hat{\eta} - \eta_0\| \xrightarrow{p} 0$, we have $\hat{\psi}_i - \varphi_i \xrightarrow{p} 0$. By the Continuous Mapping Theorem and dominated convergence (as $\hat{\psi}_i^2$ is bounded in probability), we have
$$\frac{1}{n}\sum_i T_i(\hat{\tau})(\hat{\psi}_i^2 - \varphi_i^2) \xrightarrow{p} 0.$$

Since both Term A and Term B converge to zero in probability, their sum does as well. This establishes that the numerator converges to V, and therefore $\hat{V} \xrightarrow{p} V$. □

A.8 Proof of Corollary 6.2

The proof relies on the Cramér-Wold device. We must demonstrate that for any sequence of deterministic weight vectors $c_n = (c_{1n}, \ldots, c_{K_n n})^\top \in \mathbb{R}^{K_n}$ satisfying the normalization condition $\sum_{k=1}^{K_n} c_{kn}^2 = 1$, the linear combination $c_n^\top\sqrt{n}(\hat{\theta} - \theta)$ converges in distribution to a normal distribution.

From the linear representation derived in Appendix A.7 (which establishes that $\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}}\sum_{i=1}^n T_i(\hat{\tau})\varphi_i + o_p(1)$), we can write the linear combination as:

$$c_n^\top\sqrt{n}(\hat{\theta} - \theta) = c_n^\top\left(\frac{1}{\sqrt{n}}\sum_{i=1}^n T_i(\hat{\tau})\varphi_i\right) + o_p(1).$$

Let's expand the main term:

$$\frac{1}{\sqrt{n}}\sum_{i=1}^n c_n^\top(T_i(\hat{\tau})\varphi_i) = \frac{1}{\sqrt{n}}\sum_{i=1}^n\left(\sum_{k=1}^{K_n} c_{kn}T_i(\hat{\tau})\varphi_{ik}\right).$$

Let $S_{ni} = \sum_{k=1}^{K_n} c_{kn}T_i(\hat{\tau})\varphi_{ik}$. The term S_{ni} represents the i-th component of the sum for a given n. The variables S_{ni} form a triangular array. For a fixed n, the variables $S_{n1}, S_{n2}, \ldots, S_{nn}$ are independent and identically distributed (i.i.d.) because the original observations are i.i.d. Furthermore, since $E[\varphi_{ik}|W] = 0$ (as φ_{ik} are score components centered by definition), we have $E[S_{ni}|W] = \sum_{k=1}^{K_n} c_{kn}E[T_i(\hat{\tau})\varphi_{ik}|W] = 0$. Therefore, $E[S_{ni}] = E[E[S_{ni}|W]] = 0$. Thus, the main term is a sum of i.i.d. mean-zero random variables. We will verify the Lyapunov condition for the triangular array Central Limit Theorem with a Lyapunov exponent of $\delta = 2$ (i.e., using the fourth moment). Let $\sigma_n^2 = \mathrm{Var}(S_{n1})$.

Step 1. Bounding the Variance (σ_n^2). We calculate the variance of S_{n1}:

$$\sigma_n^2 = \mathrm{Var}(S_{n1}) = E[(S_{n1})^2] - (E[S_{n1}])^2.$$

Since $E[S_{n1}] = 0$, we have:

$$\sigma_n^2 = E[(S_{n1})^2] = E\left[\left(\sum_{k=1}^{K_n} c_{kn} T_1(\hat{\tau})\varphi_{1k}\right)^2\right].$$

Expanding the square:

$$\sigma_n^2 = E\left[\sum_{k=1}^{K_n}\sum_{\ell=1}^{K_n} c_{kn} c_{\ell n} T_1(\hat{\tau})^2 \varphi_{1k}\varphi_{1\ell}\right].$$

By linearity of expectation:

$$\sigma_n^2 = \sum_{k=1}^{K_n}\sum_{\ell=1}^{K_n} c_{kn} c_{\ell n} E[T_1(\hat{\tau})^2 \varphi_{1k}\varphi_{1\ell}].$$

A crucial property of the score components is that they are orthogonal across arms for a given observation given W, i.e., $E[\varphi_{ik}\varphi_{i\ell}|W] = 0$ for $k \neq \ell$ due to the disjoint support of D_{ik} (i.e., $D_{ik}D_{i\ell} = 0$ for $k \neq \ell$). Since $T_i(\hat{\tau})$ depends on W but not directly on k or ℓ in a way that breaks this orthogonality structure, and assuming $T_i(\hat{\tau})$ is bounded and well-behaved, we can write $E[T_1(\hat{\tau})^2\varphi_{1k}\varphi_{1\ell}] = 0$ for $k \neq \ell$. Therefore, the cross-product terms vanish:

$$\sigma_n^2 = \sum_{k=1}^{K_n} c_{kn}^2 E[T_1(\hat{\tau})^2 \varphi_{1k}^2].$$

Let $V_k = E[T_1(\hat{\tau})^2\varphi_{1k}^2]$. We assume there exist finite positive constants $\underline{V}$ and $\bar{V}$ such that $0 < \underline{V} \leq V_k \leq \bar{V} < \infty$ for all k. This is a standard uniform variance bound assumption for the score components.

Then, we have:

$$\sigma_n^2 = \sum_{k=1}^{K_n} c_{kn}^2 V_k.$$

Using the bounds for V_k and the normalization condition $\sum_{k=1}^{K_n} c_{kn}^2 = 1$:

$$\underline{V}\sum_{k=1}^{K_n} c_{kn}^2 \leq \sigma_n^2 \leq \bar{V}\sum_{k=1}^{K_n} c_{kn}^2$$

$$\underline{V} \cdot 1 \leq \sigma_n^2 \leq \bar{V} \cdot 1$$

So, $\underline{V} \leq \sigma_n^2 \leq \bar{V}$, which confirms that σ_n^2 is bounded from below by a positive constant and from above by a finite constant, ensuring it does not degenerate or explode.

Step 2. Bounding the Fourth Moment ($E[S_{n1}^4]$). To apply the Lyapunov condition with $\delta = 2$, we need to bound the fourth absolute moment $E[|S_{n1}|^4] = E[S_{n1}^4]$ (since the fourth power is always non-negative).

$$E[S_{n1}^4] = E\left[\left(\sum_{k=1}^{K_n} c_{kn}T_1(\hat{\tau})\varphi_{1k}\right)^4\right].$$

We use a generalized version of Jensen's inequality (or Holder's inequality) which states that for non-negative a_k, $(\sum_{k=1}^m a_k)^p \leq m^{p-1}\sum_{k=1}^m a_k^p$. Here, $a_k = |c_{kn}T_1(\hat{\tau})\varphi_{1k}|$, $m = K_n$, and $p = 4$. Thus, for the expression inside the expectation:

$$\left(\sum_{k=1}^{K_n} c_{kn}T_1(\hat{\tau})\varphi_{1k}\right)^4 \leq \left(\sum_{k=1}^{K_n} |c_{kn}T_1(\hat{\tau})\varphi_{1k}|\right)^4 \leq K_n^{4-1}\sum_{k=1}^{K_n}(c_{kn}T_1(\hat{\tau})\varphi_{1k})^4.$$

(Note: The inequality $(\sum a_k)^4 \leq K_n^3 \sum a_k^4$ is often directly stated and used for its simplicity; it holds if a_k are not necessarily positive, but taking absolute values first is safer for rigor when using Jensen's. For even powers, this nuance is less critical for the final bound.)

Therefore,

$$E[S_{n1}^4] \leq E\left[K_n^3\sum_{k=1}^{K_n} c_{kn}^4(T_1(\hat{\tau})\varphi_{1k})^4\right].$$

By linearity of expectation:

$$E[S_{n1}^4] \leq K_n^3\sum_{k=1}^{K_n} c_{kn}^4 E[(T_1(\hat{\tau})\varphi_{1k})^4].$$

Let $M_k = E[(T_1(\hat{\tau})\varphi_{1k})^4]$. We assume there exists a finite constant C_4 such that $M_k \leq C_4$

for all k. This is a uniform fourth moment bound.

$$E[S_{n1}^4] \leq K_n^3 \sum_{k=1}^{K_n} c_{kn}^4 C_4 = C_4 K_n^3 \sum_{k=1}^{K_n} c_{kn}^4.$$

Now, we use the property that for any sequence x_k and any $p \geq q \geq 1$, $(\sum x_k^p)^{1/p} \leq (\sum x_k^q)^{1/q}$ (related to L_p norms). Specifically, $\sum_{k=1}^{K_n} c_{kn}^4 \leq (\sum_{k=1}^{K_n} c_{kn}^2)^2$. Since $\sum_{k=1}^{K_n} c_{kn}^2 = 1$, we have:

$$\sum_{k=1}^{K_n} c_{kn}^4 \leq (1)^2 = 1.$$

Substituting this back into the bound for $E[S_{n1}^4]$:

$$E[S_{n1}^4] \leq C_4 K_n^3 \cdot 1 = C_4 K_n^3.$$

Step 3. Verifying the Lyapunov Condition. The Lyapunov condition for a triangular array of i.i.d. random variables $\{S_{ni}\}_{i=1}^n$ (for each n) with mean zero and variance $\sigma_n^2 = E[S_{n1}^2]$ is satisfied if for some $\delta > 0$:

$$\lim_{n\to\infty} \frac{nE[|S_{n1}|^{2+\delta}]}{(n\sigma_n^2)^{(2+\delta)/2}} = 0.$$

We have chosen $\delta = 2$, so we need to verify:

$$\lim_{n\to\infty} \frac{nE[S_{n1}^4]}{(n\sigma_n^2)^2} = 0.$$

Let L_n denote this ratio:

$$L_n = \frac{nE[S_{n1}^4]}{n^2(\sigma_n^2)^2} = \frac{E[S_{n1}^4]}{n\sigma_n^4}.$$

Now, we substitute the bounds derived in Step 1 and Step 2: We found $E[S_{n1}^4] \leq C_4 K_n^3$ and $\sigma_n^2 \geq \underline{V}$. Therefore, $\sigma_n^4 \geq \underline{V}^2$.

$$L_n \leq \frac{C_4 K_n^3}{n\underline{V}^2}.$$

To evaluate the limit as $n \to \infty$:

$$\lim_{n\to\infty} L_n \leq \lim_{n\to\infty} \frac{C_4 K_n^3}{n\underline{V}^2} = \frac{C_4}{\underline{V}^2} \lim_{n\to\infty} \frac{K_n^3}{n}.$$

The Lyapunov condition is satisfied if this limit is equal to zero. This requires:

$$\lim_{n\to\infty} \frac{K_n^3}{n} = 0.$$

This condition is equivalent to $K_n^3 = o(n)$, or $K_n = o(n^{1/3})$. Given that the problem statement specifies $K_n = o(n^{1/3})$, this condition is indeed satisfied.

Since the Lyapunov condition holds, the Central Limit Theorem applies to the sum $\frac{1}{\sqrt{n}}\sum_{i=1}^n S_{ni}$. This means that $\frac{1}{\sqrt{n}}\sum_{i=1}^n S_{ni}$ converges in distribution to a normal distribution with mean 0 and variance $\lim_{n\to\infty}\sigma_n^2 = \lim_{n\to\infty}\sum_{k=1}^{K_n} c_{kn}^2 V_k$. By the Cramér-Wold device, the joint convergence of $\sqrt{n}(\hat{\theta}-\theta)$ to a multivariate normal distribution is established. □

A.9 Proof of Proposition 5.1

Step 1. Notation and the interacted regression. Write the saturated linear specification as

$$Y_i \;=\; \alpha_0 + \beta_0^\top W_i + \sum_{k=1}^{K} D_{ik}\left(\alpha_k + \beta_k^\top W_i\right) + \varepsilon_i, \qquad E[\varepsilon_i \mid D_i, W_i] = 0.$$

Let $r_{i\ell} = Y_i - \hat{m}_\ell(W_i) = Y_i - \hat{\alpha}_\ell - \hat{\beta}_\ell^\top W_i$ be the OLS residuals from the fully–interacted fit. Standard OLS normal equations give, *for each* $\ell = 0, \ldots, K$,

$$\sum_{i=1}^n \mathbf{1}\{D_i = \ell\}\, r_{i\ell} = 0, \qquad \sum_{i=1}^n \mathbf{1}\{D_i = \ell\}\, r_{i\ell} W_i = 0. \tag{A.1}$$

Step 2. Plug–in efficient score. With the OLS nuisance estimates and *any* probability–limit–stable $\hat{p}_\ell(W)$, the sample efficient score for arm k is

$$\hat{\psi}_{ik} \;=\; \hat{m}_k(W_i) - \hat{m}_0(W_i) + \frac{D_{ik}}{\hat{p}_k(W_i)}\, r_{ik} - \frac{D_{i0}}{\hat{p}_0(W_i)}\, r_{i0} \;-\; \theta_k.$$

Set $S_{nk} = n^{-1}\sum_{i=1}^n \hat{\psi}_{ik}$.

Step 3. Sample moment equals the interacted-OLS moment. Break S_{nk} into three pieces:

$$S_{nk} \;=\; \underbrace{n^{-1}\sum_i \big[\hat{m}_k(W_i) - \hat{m}_0(W_i)\big]}_{\text{(i)}} \;+\; \underbrace{n^{-1}\sum_i \Big[\tfrac{D_{ik}}{\hat{p}_k(W_i)}\, r_{ik} - \tfrac{D_{i0}}{\hat{p}_0(W_i)}\, r_{i0}\Big]}_{\text{(ii)}} \;-\; \theta_k.$$

Piece (ii). Fix $\ell \in \{0, k\}$, multiply and divide by the true (unknown) $p_\ell(W_i)$, and condition on W_i:

$$E\left[\frac{D_{i\ell}}{\hat{p}_\ell(W_i)}\, r_{i\ell}\middle|W_i\right] = \frac{p_\ell(W_i)}{\hat{p}_\ell(W_i)} E\Big[\mathbf{1}\{D_i = \ell\}\, r_{i\ell}\;\Big|\; W_i\Big] = 0,$$

because the inner expectation is exactly the population analogue of (A.1). Hence the law of large numbers gives $n^{-1}\sum_i D_{i\ell}\hat{p}_\ell^{-1} r_{i\ell} = o_p(1)$ for $\ell = 0, k$, so piece (ii) is $o_p(1)$.

Piece (i). By the definition of the IA estimator,

$$\hat{\theta}_k^{\mathrm{IA}} \;=\; n^{-1}\sum_i \big[\hat{m}_k(W_i) - \hat{m}_0(W_i)\big].$$

Therefore $S_{nk} = \hat{\theta}_k^{\mathrm{IA}} - \theta_k + o_p(1)$. Imposing the sample moment condition $S_{nk} = 0$ yields $\hat{\theta}_k^{\mathrm{IA}} = \theta_k + o_p(1)$, *exactly the solution to the OLS normal equations.* Thus the plug–in DML estimator with the saturated linear outcome model returns the interacted-ATE estimator (Goldsmith-Pinkham *et al.*, 2024). Since (4) is Neyman-orthogonal, $\hat{\theta}_k$ inherits all DML asymptotics. □

A.10 Proof of Proposition ??

In the restricted subsample $\{D = 0\} \cup \{D = k\}$ let $q_\ell(W) = P(D = \ell \mid W, D \in \{0, k\})$ be the *binary* propensity scores. Then $\varphi_k^\star$ becomes

$$\psi_i^{(EW)} \;=\; \big[m_k(W_i) - m_0(W_i)\big] + \frac{D_{ik}}{q_k(W_i)}\{Y_i - m_k(W_i)\} - \frac{D_{i0}}{q_0(W_i)}\{Y_i - m_0(W_i)\} - \theta_k.$$

This is the standard AIPW score for a single treatment arm, whose expectation is zero at the truth by the classical proof in Chernozhukov *et al.* (2018). □

A.11 Proof of Proposition 5.2

Step 1. CW-weighted propensities. Let $\pi_\ell = P(D = \ell)$ and write $g(W) = \Big[\sum_{\ell=0}^K \pi_\ell(1 - \pi_\ell)/p_\ell(W)\Big]^{-1}$. Define the *CW-weighted indicators* $\tilde{D}_{i\ell} = g(W_i)\, D_{i\ell}/p_\ell(W_i)$. By construction they satisfy $E[\tilde{D}_{i\ell} \mid W] = g(W)$ and hence $E[\tilde{D}_{i\ell}] = 1$.

Step 2. GPHK-CW estimating equation. (Goldsmith-Pinkham *et al.*, 2024) estimate θ_k as the coefficient from the weighted regression of Y on D_{ik} using weights $g(W_i)$. The corresponding sample normal equation is

$$0 \;=\; \sum_{i=1}^{n} g(W_i)(D_{ik} - \pi_k)\big(Y_i - \hat{\theta}_k^{\mathrm{CW}}\big). \tag{A.3}$$

Step 3. Plug–in score equals the CW moment. Insert the CW propensities $\hat{p}_\ell(W) = D_{i\ell}/\tilde{D}_{i\ell}$ into the efficient score:

$$\hat{\psi}_{ik}^{\mathrm{CW}} = \big[\hat{m}_k(W_i) - \hat{m}_0(W_i)\big] + \tilde{D}_{ik}\{Y_i - \hat{m}_k(W_i)\} - \tilde{D}_{i0}\{Y_i - \hat{m}_0(W_i)\} - \theta_k.$$

Average and rearrange,

$$\begin{aligned} 0 \overset{!}{=} & \sum_i \hat{\psi}_{ik}^{\mathrm{CW}} \\ = & \sum_i \tilde{D}_{ik} Y_i \;-\; \sum_i \tilde{D}_{i0} Y_i \;-\; n\theta_k. \end{aligned}$$

Use (A.2) with $\ell = 0, k$ to replace $\sum_i \tilde{D}_{i0} = \sum_i \tilde{D}_{ik} = n$, giving $\sum_i g(W_i)(D_{ik} - \pi_k)Y_i = n\theta_k$. Subtracting $\theta_k \sum_i g(W_i)(D_{ik} - \pi_k) = 0$ from both sides we obtain the GPHK-CW normal equation (A.3). Hence the plug-in solution $\hat{\theta}_k^{\mathrm{CW}}$ is *identical* to the common-weight estimator, and the score expectation is zero at the truth. Orthogonality follows because (4) is orthogonal for any propensities. □

A.12 Rate Conditions for Common Learners

Random forests with honesty, boosted trees of depth $O(\log n)$, gradient-boosted splines, and ReLU networks of depth $\le C \log n$ achieve the $n^{-1/4}$ L_2 rate under standard sparsity or smoothness conditions; see Athey and Wager (2018), Farrell *et al.* (2021).

A.13 Auxiliary lemmas for the MSE–selected threshold

Lemma A.6 (Consistency of the MSE-minimising threshold)**.** *Let $\mathcal{G} = \{\tau_1 < \cdots < \tau_G\}$ be a finite grid and define $\widehat{\mathrm{MSE}}(\tau) = \widehat{V}(\tau) + \widehat{B}(\tau)^2$ as in Section 6.2. Suppose*

$$\sup_{\tau \in \mathcal{G}} \big| \widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau) \big| = o_p(1),$$

and that the population MSE has a unique minimiser $\tau^\star = \arg\min_{\tau \in \mathcal{G}} \mathrm{MSE}(\tau)$. Then the empirical minimiser $\hat{\tau} = \arg\min_{\tau \in \mathcal{G}} \widehat{\mathrm{MSE}}(\tau)$ satisfies $\hat{\tau} \to_p \tau^\star$.

Proof. Let the finite grid be $\mathcal{G} = \{\tau_1 < \cdots < \tau_G\}$ and recall the decompositions

$$\mathrm{MSE}(\tau) = V(\tau) + B(\tau)^2, \qquad \widehat{\mathrm{MSE}}(\tau) = \widehat{V}(\tau) + \widehat{B}(\tau)^2.$$

By hypothesis

$$\sup_{\tau \in \mathcal{G}} \big| \widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau) \big| = o_p(1). \tag{A.1}$$

Step 1. A strictly positive population gap. Because $\tau^\star$ is the unique minimizer of MSE over $\mathcal{G}$ (an assumption in the lemma), define the gap

$$\Delta \;=\; \min_{\tau \in \mathcal{G} \setminus \{\tau^\star\}} \big\{ \mathrm{MSE}(\tau) - \mathrm{MSE}(\tau^\star) \big\} \;>\; 0. \tag{A.2}$$

Step 2. Bounding the mis-selection probability. If the data-driven choice satisfies $\hat{\tau} \neq \tau^\star$, then by definition of $\hat{\tau}$, $\widehat{\mathrm{MSE}}(\hat{\tau}) \leq \widehat{\mathrm{MSE}}(\tau^\star)$. Therefore

$$\begin{aligned}
\Delta &\leq \mathrm{MSE}(\hat{\tau}) - \mathrm{MSE}(\tau^\star) \\
&\leq \big| \widehat{\mathrm{MSE}}(\hat{\tau}) - \mathrm{MSE}(\hat{\tau}) \big| + \big| \widehat{\mathrm{MSE}}(\tau^\star) - \mathrm{MSE}(\tau^\star) \big| \\
&\leq 2 \sup_{\tau \in \mathcal{G}} \big| \widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau) \big|.
\end{aligned}$$

Thus

$$\{\hat{\tau} \neq \tau^\star\} \;\subseteq\; \Big\{ \sup_{\tau \in \mathcal{G}} \big| \widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau) \big| > \tfrac{\Delta}{2} \Big\}.$$

Step 3. Conclude via uniform consistency. Taking probabilities and invoking (A.1) yields

$$P(\hat{\tau} \neq \tau^\star) \;\leq\; P\Big(\sup_{\tau \in \mathcal{G}} \big| \widehat{\mathrm{MSE}}(\tau) - \mathrm{MSE}(\tau) \big| > \tfrac{\Delta}{2} \Big) \;=\; o(1).$$

Hence $\hat{\tau} \xrightarrow{p} \tau^\star$, and—because $\mathcal{G}$ is finite—the stochastic-equicontinuity arguments used in Section 6 hold uniformly on $\mathcal{G}$. □ □

Lemma A.7 (Uniform consistency of $\widehat{\mathrm{MSE}}$)**.** *If Assumptions 2.6 and 2.4 hold and $\mathcal{G}$ is finite, then*

$$\sup_{\tau \in \mathcal{G}} \big| \widehat{V}(\tau) - V(\tau) \big| \;=\; o_p(1), \qquad \sup_{\tau \in \mathcal{G}} \big| \widehat{B}(\tau) - B(\tau) \big| \;=\; o_p(1).$$

Proof. Write $\varphi_i = \varphi(Z_i; \theta, \eta)$ for the true influence function evaluated at the true nuisances. For every $\tau \in \mathcal{G}$, define

$$\widehat{V}(\tau) = \frac{1}{n} \sum_{i=1}^{n} T_i(\tau)\, \varphi_i^2, \quad V(\tau) = E\big[T_i(\tau)\, \varphi_i^2\big],$$

$$\widehat{B}(\tau) = \frac{1}{n}\sum_{i=1}^{n} T_i(\tau)\,\varphi_i, \quad B(\tau) = E\big[T_i(\tau)\,\varphi_i\big].$$

Step 1. Uniform LLN for $\widehat{V}$. The function class

$$\mathcal{F}_V \;=\; \big\{\, f_\tau(Z) = T(\tau)\varphi(Z)^2 : \tau \in \mathcal{G}\big\}$$

contains exactly G measurable elements. Because trimming enforces $T(\tau) \le 1$ and $E[\varphi(Z)^2] < \infty$ (square-integrability of the score), the envelope $F_V(Z) = \varphi(Z)^2$ is integrable.

For a finite class of integrable functions, the finite-class law-of-large-numbers (Van der Vaart and Wellner, 1996) gives

$$\sup_{\tau\in\mathcal{G}}\big|\widehat{V}(\tau) - V(\tau)\big| \;=\; o_p(1). \tag{B.1}$$

Step 2. Uniform LLN for $\widehat{B}$. Apply the same reasoning to

$$\mathcal{F}_B \;=\; \big\{\, g_\tau(Z) = T(\tau)\varphi(Z) : \tau \in \mathcal{G}\big\},$$

whose integrable envelope is $F_B(Z) = |\varphi(Z)|$. Then

$$\sup_{\tau\in\mathcal{G}}\big|\widehat{B}(\tau) - B(\tau)\big| \;=\; o_p(1). \tag{B.2}$$

Moreover, $\sup_{\tau\in\mathcal{G}}|B(\tau)| \le E[|\varphi(Z)|] < \infty$, so $\max_\tau |B(\tau)|$ is a fixed finite constant; similarly, $\sup_\tau |\widehat{B}(\tau)| = O_p(1)$ by the triangle inequality and (B.2).

Step 3. Uniform consistency of $\widehat{\text{MSE}}$. For each $\tau \in \mathcal{G}$,

$$\begin{aligned}\big|\widehat{\text{MSE}}(\tau) - \text{MSE}(\tau)\big| &= \big|\widehat{V}(\tau) - V(\tau)\big| \;+\; \big|\widehat{B}(\tau)^2 - B(\tau)^2\big| \\ &\le \big|\widehat{V}(\tau) - V(\tau)\big| \;+\; 2\,\big|\widehat{B}(\tau) - B(\tau)\big|\,\big\{|\widehat{B}(\tau)| + |B(\tau)|\big\}.\end{aligned}$$

Take the supremum over $\tau \in \mathcal{G}$ and apply (B.1)–(B.2):

$$\sup_{\tau\in\mathcal{G}}\big|\widehat{\text{MSE}}(\tau) - \text{MSE}(\tau)\big| \;\le\; o_p(1) + 2\,o_p(1)\,O_p(1) \;=\; o_p(1).$$

The two statements of the lemma, (B.1) and (B.2), have now been proved. □

A.14 Proof of Theorem 6.3

Let clusters be indexed by $g = 1, \ldots, G$, with (random) cluster size n_g and observations $Z_{gi} = (Y_{gi}, D_{gi}, W_{gi})$, $i = 1, \ldots, n_g$. Write $N = \sum_{g=1}^{G} n_g$ and denote by $\hat{\eta}^{(-j)}$ the nuisance estimates trained on clusters not in fold j (cluster-level cross-fitting). For a deterministic threshold $\tau > 0$ define

$$T_{gi}(\tau) = \mathbf{1}\{\min_k p_k(W_{gi}) \geq \tau\} \quad (\text{or } \mathbf{1}\{f(D_{gi} \mid W_{gi}) \geq \tau\} \text{ in the continuous case}).$$

Let the *raw* moment be $\psi(Z;\eta)$ (no $-\theta$) and $\varphi(Z) = \psi(Z;\eta_0) - \theta$ the population influence function at the truth η_0, as in (4)–(3). Define cluster totals

$$S_g(\tau) \;=\; \sum_{i=1}^{n_g} T_{gi}(\tau)\,\{\psi(Z_{gi};\eta_0) - \theta(\tau)\}, \qquad C_g(\tau) \;=\; \sum_{i=1}^{n_g} T_{gi}(\tau),$$

and their expectations $\mu_C(\tau) = E[C_g(\tau)]$ and $\sigma_S^2(\tau) = E[S_g(\tau)^2]$.[1]

Step 1. The estimator computed by Algorithm 1 can be written as the ratio of cluster sums of trimmed raw moments (recall we average the raw moment to form $\hat{\theta}$):

$$\hat{\theta}(\tau) = \frac{\sum_{g=1}^{G}\sum_{i=1}^{n_g} T_{gi}(\tau)\,\hat{\psi}_{gi}}{\sum_{g=1}^{G}\sum_{i=1}^{n_g} T_{gi}(\tau)}, \qquad \hat{\psi}_{gi} = \psi\big(Z_{gi};\hat{\eta}^{(-j(g))}\big).$$

Add and subtract $\theta(\tau)$ in the numerator, and use the identity $\hat{\psi}_{gi} - \theta(\tau) = \{\psi(Z_{gi};\eta_0) - \theta(\tau)\} + \{\hat{\psi}_{gi} - \psi(Z_{gi};\eta_0)\}$ to obtain

$$\begin{aligned}\hat{\theta}(\tau) - \theta(\tau) &= \frac{\sum_{g=1}^{G} S_g(\tau)}{\sum_{g=1}^{G} C_g(\tau)} \;+\; \frac{\sum_{g=1}^{G}\sum_{i=1}^{n_g} T_{gi}(\tau)\,\{\hat{\psi}_{gi} - \psi(Z_{gi};\eta_0)\}}{\sum_{g=1}^{G} C_g(\tau)} \\ &=: \frac{\sum_{g=1}^{G} S_g(\tau)}{\sum_{g=1}^{G} C_g(\tau)} \;+\; R_G(\tau). \end{aligned} \tag{16}$$

Under Assumptions 2.6 and 6.1 (orthogonality, $N^{-1/4}$ L_2 rates on the trimmed support, cluster-level cross-fitting, and $E[n_g] \in (0,\infty)$), the *nuisance remainder* satisfies

$$\sqrt{G}\,R_G(\tau) \;=\; \frac{1}{\sqrt{G}}\sum_{g=1}^{G}\sum_{i=1}^{n_g} T_{gi}(\tau)\big\{\hat{\psi}_{gi} - \psi(Z_{gi};\eta_0)\big\}\,\Big/\Big(\tfrac{1}{G}\sum_{g=1}^{G} C_g(\tau)\Big) \;=\; o_p(1).$$

The proof mirrors Lemma A.4: expand the difference by a first-order Gateaux derivative, note that the expectations of the linear terms vanish by Neyman orthogonality (Lemmas 3.1–3.2), and bound the quadratic terms by $\sqrt{G} \times o_p(N^{-1/2}) = o_p(1)$ because $N/G \to_p E[n_g] \in (0,\infty)$.

[1] By definition of $\theta(\tau) = E[\psi(Z;\eta_0) \mid T(\tau) = 1]$, we have $E[S_g(\tau)] = 0$.

Step 2. By a cluster LLN and $\mu_C(\tau) = E[C_g(\tau)] > 0$,

$$\frac{1}{G}\sum_{g=1}^{G} C_g(\tau) \ \xrightarrow{p}\ \mu_C(\tau), \qquad \sqrt{G}\Big\{\frac{1}{G}\sum_{g=1}^{G} C_g(\tau) - \mu_C(\tau)\Big\} = O_p(1).$$

Substituting this into (16) and multiplying by $\sqrt{G}$ yields

$$\sqrt{G}\left\{\hat{\theta}(\tau) - \theta(\tau)\right\} = \frac{1}{\mu_C(\tau)} \cdot \frac{1}{\sqrt{G}}\sum_{g=1}^{G} S_g(\tau) \ + \ o_p(1), \tag{17}$$

because $E[S_g(\tau)] = 0$ makes the usual ratio correction term vanish.

Step 3. Clusters are independent by Assumption 6.1. Moreover, $E[S_g(\tau)] = 0$ and, by Lemma A.2 combined with $E[n_g^{2+\kappa}] < \infty$, we have $E[|S_g(\tau)|^{2+\kappa}] \le C < \infty$ for some $\kappa > 0$. Hence the Lindeberg–Feller CLT for independent, not-necessarily identically distributed arrays applies:

$$\frac{1}{\sqrt{G}}\sum_{g=1}^{G} S_g(\tau) \ \xrightarrow{d}\ N\big(0,\ \sigma_S^2(\tau)\big), \qquad \sigma_S^2(\tau) = E\big[S_g(\tau)^2\big].$$

Combining with (17) gives

$$\sqrt{G}\left\{\hat{\theta}(\tau) - \theta(\tau)\right\} \ \xrightarrow{d}\ N\Big(0,\ V_{\text{cl}}(\tau)\Big), \qquad V_{\text{cl}}(\tau) = \frac{\sigma_S^2(\tau)}{\mu_C(\tau)^2}.$$

Step 4. Let $\mathcal{G}_n$ be the finite trimming grid in Assumption 6.1, and let $\hat{\tau}$ minimize the uniform MSE proxy in Section 6.2. The cluster versions of Lemmas A.3–A.4 (finite-class LLN for $\{S_g(\tau), C_g(\tau) : \tau \in \mathcal{G}_n\}$ and the same orthogonality argument) imply

$$\hat{\tau} \ \xrightarrow{p}\ \tau^\star \in \mathcal{G}_\infty, \quad \text{and} \quad \sup_{\tau \in \mathcal{G}_n}\left|\sqrt{G}\left\{\hat{\theta}(\tau) - \theta(\tau)\right\} - \frac{1}{\mu_C(\tau)} \cdot \frac{1}{\sqrt{G}}\sum_{g=1}^{G} S_g(\tau)\right| = o_p(1).$$

Therefore,

$$\sqrt{G}\left\{\hat{\theta}(\hat{\tau}) - \theta(\tau^\star)\right\} \ \xrightarrow{d}\ N\Big(0,\ V_{\text{cl}}(\tau^\star)\Big).$$

If the grid vanishes (Condition 6.1) so that $\tau^\star \to 0$ and the trimming bias is $o_p(G^{-1/2})$, then $\theta(\tau^\star) \to \theta$ and the limit variance equals $V_{\text{cl}}(0)$.

Step 5. Define the cluster sums of estimated centered contributions

$$\hat{S}_g = \sum_{i=1}^{n_g} T_{gi}(\hat{\tau}) \{\hat{\psi}_{gi} - \hat{\theta}\}, \qquad \hat{\mu}_C = \frac{1}{G}\sum_{g=1}^{G} C_g(\hat{\tau}).$$

The proposed estimator

$$\hat{V}_{\text{cl}} = \frac{G}{\left(\sum_g C_g(\hat{\tau})\right)^2} \sum_{g=1}^{G} \hat{S}_g^2 = \frac{1}{\hat{\mu}_C^2} \cdot \frac{1}{G} \sum_{g=1}^{G} \hat{S}_g^2$$

converges in probability to $V_{\text{cl}}(\tau^\star)$ because (i) $\hat{\mu}_C \to_p \mu_C(\tau^\star)$ by a cluster LLN, (ii) $\frac{1}{G}\sum_g \hat{S}_g^2 \to_p E[S_g(\tau^\star)^2]$ (the difference $\hat{S}_g - S_g(\tau^\star)$ is $o_p(1)$ uniformly over g by the same orthogonality and rate arguments), and (iii) $\mathcal{G}_n$ is finite. A small-sample degrees-of-freedom factor $G/(G-1)$ may be multiplied if desired.

Putting Steps 1–5 together proves the stated clustered CLT and the consistency of the cluster-robust variance estimator under both deterministic and data-driven trimming. The argument holds verbatim for the continuous-dose score (3) and the discrete multi-arm score (4).